Ask for Fish, Find the Hagfish: A VLM Names What a Promptable Segmenter Cannot See, for Abundance Counts in Baited Underwater Video

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Abstract. Baited underwater video surveys report abundance as MaxN, the largest number of individuals of a species visible in one frame. SAM 3.1, a promptable segmenter, tracks whatever short noun it is given, and only that: on our 39 clips “animal” returned no box on 29, and the name “luderick” returned nothing on all six clips of that species. Without training, we put a vision-language model (VLM), Muse Spark, in front of it. The VLM names animals the request left out, in nouns the segmenter answers; where no word works, it picks moving spots that keep their shape, and the segmenter tracks each from a box with no word. The VLM then audits every track, and doubtful ones go to a person instead of the count. Discovery is scored as presence of taxa. On four development nautilus clips the pipeline finds all seven taxa a person listed; on a night clip where 17 fixed words box one of four nautiluses, it counts all four. On five held-out videos it names all four non-fish animals marked beforehand, counts three, queues the fourth and counts nothing that is not an animal; on six more it counts the one animal two people marked beforehand. For the requested species, on 24 held-out clips, the pipeline’s MaxN is closer to a hand count of the videos (by the first author, 21 of 24 counted blind) than the reference boxes are (summed error 6 against 16, within one fish on all 24), and none of 40 randomly drawn boxes scored as false positives was judged not a fish: corrected precision is 93.7–100%, against 77.2% on the boxes. The audits keep the segmenter’s recall and match a tuned track-length rule. As an auditor the VLM adopts names placed in its prompt. The result is an annotation accelerator whose output a person confirms.

Claims and evidence. n = clips.

claim	evidence	n	caveat
SAM 3.1 answers narrow words: “animal” gives no box on 29 of 39 clips; “luderick” gives nothing on all six clips of that species	§7.4, Table 6	39; 6	our clips and one model version; not a property of SAM 3.1 in general
On luderick the VLM audits match a tuned minimum-track-length rule and cannot raise box recall above the segmenter’s	§6.1 Table 1; §6.2 Table 1c	6 development; 24 held-out	one estuary; the length threshold was chosen with test labels on the development clips; MaxN-exact cannot separate methods (always predicting the most common value is exact on 5 of 6 and 17 of 24)
Against a hand count of all 24 held-out videos the pipeline’s MaxN is exact on 18, within ±1 on 24, summed error 6; the reference boxes 16, 22 and 16	§6.2 Table 1d	24 held-out	one counter, the first author; 21 counted blind and shuffled, 3 after being flagged as outliers
One person fixes the held-out reference in 9 min 39 s: 20 tracks judged on a phone, 15 unboxed fish (145 boxes) added, 0 not a fish; the corrected MaxN matches the hand count as closely as the pipeline (Σ|err| 6 vs 16)	§6.2	24 held-out; 20 tracks + 3 controls	added boxes come from the pipeline, so the pipeline is not re-scored against them; the judge had seen 40 of the boxes before; re-judged, 11 of 11 earlier “fish” tracks were fish again
Most held-out “false positives” are unboxed fish: of 40 drawn at random, 29 fish, 2 part of a fish, 9 unsure, 0 not a fish (precision 93.7–100% vs 77.2% scored)	§6.2	40 boxes; 10 controls	one judge, labels hidden; 10/10 matched controls called fish
The person queue holds most tracks behind the largest over-counts, and also real fish	§6.1 Table 1b; §6.2	24 held-out	queue is 15 of 49 tracks; rejecting it costs 3.3 points of recall
Clearing the queue is quick: 9 queued tracks of four clips took one person 121 s on a phone; all 6 accepted tracks overlap the human boxes, the 3 left unsure match none	§6.1	4 development; 9 tracks	one judge, one sitting; development clips
Asked to count each frame itself, the VLM’s MaxN is above the human box count on 16 of 24 held-out clips	§6.2, Table 1c	24; 6	no locations, so a count cannot be checked; some “excess” may be unboxed fish
On nautilus clips VLM naming finds six of seven	§6.3, Tables 2–3	4 development; 7 taxa	development data; taxa scored as present, not

claim	evidence	n	caveat
taxa without any audit (seven with a motion cue); audits fix names, duplicates and one false taxon			counts; the reference is a non-expert's list from four frames
On five held-out videos the pipeline names 4 of 4 non-fish animals marked beforehand, counts 3, queues 1, counts no non-animal	§6.3, Table 3b	5; 4 animals	clips chosen by a rule that favours non-fish animals; presence only; "not an animal" judged by the author with labels hidden
On six more held-out videos it counts the one animal marked beforehand and no non-animal	§6.3, Table 3c	6; 1 animal	four of the six were chosen because their source text names a non-fish animal; the fixed 16-word baseline finds the same animals
The VLM auditor adopts names in its prompt and changes verdicts with the layout of the evidence	§7.1, §7.5	6 development	development clips; repeated calls also vary on their own
Muse Wonder, Meta's larger VLM, given Spark's 101 questions, rejects tiles of a crab a person was certain of, gives 12 unsupported names against Spark's 5, and costs about 4.3× per call	§7.6, Table 7	15; 101 questions	same questions, not Wonder's own run; yes/no per tile without confidence thresholds
A plain propose–segment–look–revise loop (the SAM 3 Agent pattern) with the same models reaches 3 of 5 held-out marked animals and 5 of 7 development taxa, against 4 of 5 and 7 of 7	§7.7, Table 8	15	text prompts, full frame, forward only; the published agent's prompts are not reproduced
An open-vocabulary detector (OWLv2) given "animal" and the same 16 words overlaps 39 of the 45 counted non-fish tracks, but none through "animal", 17 only as "fish" and 11 with a word of the right group, and adds 1,716 unmatched non-fish boxes	§7.8, Table 9	15	one detector, base size, no tracking; the reference is what the pipeline counted, so it measures agreement, not recall

1 Introduction

A baited remote underwater video system (BRUV) is a camera lowered next to a bait bag; it records for an hour or more while animals arrive, feed and leave. The standard abundance metric extracted from such footage is MaxN: for each species, the largest number of individuals visible in one frame (Priede et al., 1994; see Langlois et al., 2020). MaxN is conservative

by construction, since an animal that leaves and returns can never be counted twice, and it is "the most commonly reported metric" for BRUVs (Whitmarsh et al., 2017, as cited by Connolly et al., 2021). It is also expensive. BRUV analysis "typically tak[es] several hours per hour of video" (Francour et al., 1999, as cited by Marrable et al., 2022), and field guides ask annotators to identify "all fish … to the lowest taxonomic level possible" (Langlois et al., 2020, §2.6.3), then have a senior analyst re-check 10% of videos against a 95% accuracy bar (§2.6.7).

Automated counting for BRUV has made real progress, but almost always for a fixed list. Detectors have been trained for luderick (Ditria et al., 2020a), Australasian snapper (Connolly et al., 2021), yellowfin bream (Lopez-Marcano et al., 2021), elasmobranchs (Varini et al., 2024) and up to twelve target species plus a catch-all "fish" class (Marrable et al., 2022). Single-habitat detectors transfer poorly to other habitats (Ditria et al., 2020b). Two properties of the manual task are therefore hard to reach. First, the species list is open: a nautilus survey also films hagfish and shrimp, as the supplementary video captions of Barord et al. (2014) note, and those animals are often the interesting part. A detector trained on one taxon has no way to report them. Second, a count is only useful if a scientist can check it. Every number needs an evidence frame, and doubtful cases need to be surfaced rather than silently resolved.

Promptable foundation models change what an off-the-shelf system can attempt. SAM 3 performs promptable concept segmentation in video, returning masks and identities "for all matching object instances" of a short noun phrase (Carion et al., 2026); SAM 3.1 adds faster multi-object tracking (Meta, 2026). A general VLM can describe a frame in open vocabulary. Neither alone does the BRUV task. The segmenter tracks only what it is told to look for, and we show that the words it responds to are narrow. On still frames the VLM names animals, but it returns no identities to follow through a clip, and as a judge it is subject to the anchoring, sycophancy and order effects documented for language and vision-language models (Jones and Steinhardt, 2022; Dang et al., 2024; Tian et al., 2025).

This paper is an analysis of combining the two for abundance counting when the species list is not fixed in advance, organised around measured failure modes rather than around a model. Our contributions are:

1. **Measurements of what a promptable video segmenter responds to on BRUV footage.** On 29 of our 39 clips SAM 3.1 returned nothing for the hypernym "animal", and on all six luderick development clips nothing for "luderick"; a rare name failed where a basic-level neighbour worked ("hagfish" vs. "eel"); word-prompted tracks began only once an animal was clear (a reversed pass recovered 16 → 60 frames for one nautilus); and a box prompt with no word tracked an animal that no word or zoom ever found (§7.4).
2. **A pipeline that uses the VLM as namer, translator and auditor of the segmenter,** with three audits aimed at the ways a MaxN goes wrong (a false track, a duplicated animal in the peak frame, a wrong box in the peak frame), a motion-residual path for animals no noun reaches, and a second unprimed look for tracks that rest on weak evidence (§4).
3. **An evaluation on two tasks** (counting against human boxes, discovery of untargeted taxa) **against model-free and no-audit baselines, with a design ablation** that shows recall and MaxN accuracy moving with the layout of the evidence sheet, including a variant that scores better on MaxN for the wrong reason, and repeated calls that separate the

layout effect from nondeterminism (§6, §7.1). The baselines also bound the claim: on single-category counting a track-length rule does as well as the audits on the development clips and only one held-out clip worse, and on discovery the naming step, not the audits, finds the taxa.

4. **A catalogue of VLM-auditor failure modes observed in a counting pipeline**, each tied to the count it changed: name anchoring, vocabulary priming, layout and context sensitivity, nondeterminism, wrong rejections in the peak check, malformed output, and species names that are only proposals (§7.5).

We position the auditing loop as an application of a known pattern rather than a new agent: SAM 3's own data engine uses MLLMs as "AI verifiers" of masks, and its "SAM 3 Agent" lets an MLLM propose noun phrases and check the returned masks (Carion et al., 2026, §4 and §6). What is new here is the target (ecological counts with per-number evidence), the specific checks that a MaxN requires, and measurements of how those checks fail.



Figure 1. The same seagrass frame (luderick archive, clip 20190618_9, frame 35; CC BY 4.0) three ways. Left: SAM 3.1 prompted with "animal", 0 detections (none in any of the clip's 107 frames). Middle: prompted with "fish", 10 detections. Right: the audited pipeline, with the names the VLM audit gave each track (7 striped seabream, 3 fish left unidentified).

2 Related work

MaxN and BRUV annotation. MaxN, also defined as MAXNO for baited video by Ellis and DeMartini (1995), is the field's standard abundance index (Langlois et al., 2020). It is known to be a non-linear, saturating function of true abundance (Schobernd et al., 2014) and to depend on non-density factors such as swimming speed and bait plume (Kilfoil et al., 2021); Ellis and DeMartini (1995) already noted earlier deep-sea work in which MAXNO was unrelated or inversely related to density. We do not take a position on MaxN as an index; we take it as the quantity annotators are asked to produce, and note that because it is a single-frame maximum it inherits any duplicate or spurious detection in that one frame (Connolly et al., 2021, report raw detector counts off by up to 50% at high density). Shortcuts already used to cut effort have costs: subsampling 40 frames per hour kept MeanCount accurate but underestimated species richness by about 40%, "missing the less common species" (Hsu et al., 2025).

Automated fish detection and counting. Closed-set detectors with tracking or sequence suppression are the norm (Ditria et al., 2020a; Lopez-Marcano et al., 2021; Connolly et al., 2021; Varini et al., 2024). Marrable et al. (2022) is the broadest BRUV system we found, and still reports that "88% of fish were located and given a fish label and identified for manual labelling". Motion has been used to generate fish candidate regions for training a detector (Salman et al., 2020) and background modelling with a false-positive filter is a classic approach for fixed underwater cameras (Giordano et al., 2016). Datasets such as DeepFish

(Saleh et al., 2020), FishNet (Khan et al., 2023) and the luderick seagrass set (Ditria et al., 2021) support this line. Marine segmentation foundation models are typically fine-tuned: CoralSCOP for coral stills (Zheng et al., 2024b), Dual-SAM for single-image marine animals (Zhang et al., 2024), SAM-U for underwater video after a reported 13-point J&F drop of open-air models (Zhao et al., 2026). SAM-family models have been used on BRUV footage in a training-free pipeline (Kavasidis et al., 2025) and as an annotation aid (Sánchez et al., 2026). To our knowledge no prior BRUV work names and counts untargeted taxa without a species list, or uses a VLM to audit a tracker's output at the MaxN frame.

Open-set recognition in the ocean. FathomNet aggregates expert-labelled ocean imagery for 2,244 concepts (Katija et al., 2022), and its 2023 competition frames recognising "a new organism, an unusual collection of animals, or … out-of-sample" data as central (Orenstein et al., 2023). These works evaluate identification on images; they do not connect open-vocabulary naming to an abundance metric and its evidence frame.

Promptable and open-vocabulary segmentation. SAM (Kirillov et al., 2023) and SAM 2 (Ravi et al., 2025) made segmentation promptable by points and boxes; SAM 2 notes failures "in crowded scenes, after long occlusions or in extended videos". SAM 3 adds concept prompts, constrained to "simple noun phrases" for "atomic visual concepts", and lists weak zero-shot generalisation "to fine-grained out-of-domain concepts" as a limitation (Carion et al., 2026). Open-vocabulary detectors (Liu et al., 2024; Minderer et al., 2022, 2023; Cheng et al., 2024) show the same pattern on benchmarks: OVDEval reports that top models "all fail on the new tasks except for simple object types" (Yao et al., 2024), detection depends on which synonym names an object (Nebbia and Kovashka, 2024) and on the choice of class names (Parisot et al., 2023). VLMs themselves prefer basic-level categories in the sense of Rosch et al. (1976) (Sawyer et al., 2025, for two open VLMs other than the one used here), which suggests that a VLM can act as the translator between a scientist's request and the segmenter's vocabulary. Underwater, SAM 2's quality is "extremely dependent on the type of user-provided prompts", with ground-truth boxes working far better than automatic point prompts (Lian and Li, 2024).

VLMs as verifiers, and how they fail. LLMs and MLLMs are used as judges and critics (Zheng et al., 2023; Chen et al., 2024; Xiong et al., 2025), as tool users that call vision models (Gupta and Kembhavi, 2023; Surís et al., 2023) and as reasoning segmenters (Lai et al., 2024). Marks drawn on an image steer a VLM's attention (Yang et al., 2023; Shtedritski et al., 2023), and cropping helps with small subjects (Zhang et al., 2025). Their failure modes are well documented: object hallucination (Rohrbach et al., 2018; Li et al., 2023), sycophancy (Sharma et al., 2024; Li et al., 2025), flips after a single misleading cue in 65% of cases (Dang et al., 2024), anchoring that is not fixed by telling the model to ignore the anchor (Lou and Sun, 2024), sensitivity to option order (Pezeshkpour and Hruschka, 2024; Zheng et al., 2024a) and to image position in multi-image inputs (Tian et al., 2025), and run-to-run nondeterminism even at temperature zero (Atil et al., 2024; He, 2025). Self-correction without external feedback is unreliable (Huang et al., 2024); answering verification questions independently, so that "answers are not biased by other responses", reduces hallucination (Dhuliawala et al., 2023). Our second unprimed look follows that idea.

Humans in the loop. Camera-trap work combines models with people: automation is gated by confidence with the rest sent to people (Norouzzadeh et al., 2018; Beery et al., 2019),

models trained on citizen-science labels reduce human effort on new studies (Willi et al., 2019), and accuracy drops on faunas new to a model (Tabak et al., 2019). Marine annotation platforms such as BIIGLE (Langenkämper et al., 2017), MAIA (Zurowietz et al., 2018), CoralNet (Beijbom et al., 2015) and Squidle+ (Friedman et al., 2026) keep a person in charge with machine suggestions and "Needs Review" flags. Verification instead of drawing is a long-standing way to cut annotation cost (Papadopoulos et al., 2016). We follow this norm: the pipeline's output is a set of counts plus a queue of tracks that a person should confirm.

3 Problem setting

Input. A clip $V = (I_1, \ldots, I_T)$ from a fixed camera, re-encoded to 5 frames per second, and a request R: a list of plain nouns (in this paper always $R = \{\mathsf{fish}\}$).

Output. A set of tracks $\mathcal{T}$, each with frames $F_k \subseteq \{1..T\}$, a box $b_{k,t}$ per frame, a label ℓ_k (a species name, a category marked "unidentified", or *none* if rejected), and a flag q_k meaning "a person should look at this". For each label s,

$$\mathsf{MaxN}_s = \max_t |\{k : \ell_k = s,\ t \in F_k\}|, \qquad t^\star_s = \operatorname*{argmax}_t |\{k : \ell_k = s,\ t \in F_k\}|,$$

with the evidence frame $t^\star_s$ rendered with every counted mask, and the time of first arrival $\min_k \min F_k$ over tracks labelled s. Labels are not restricted to R: an animal outside the request is reported under its own name.

What a person does with it. The reviewer confirms or corrects the queued tracks and spot-checks the evidence frames. A count is "final" only after that review; the measurements below are of the pipeline output *before* review, together with the size of the queue that review would require.

4 Method

Figure 2 shows the pipeline. The design principle is that each model does only what it was measured to do well, and that every decision which can change a count is made on evidence a person can inspect afterwards (a saved sheet).

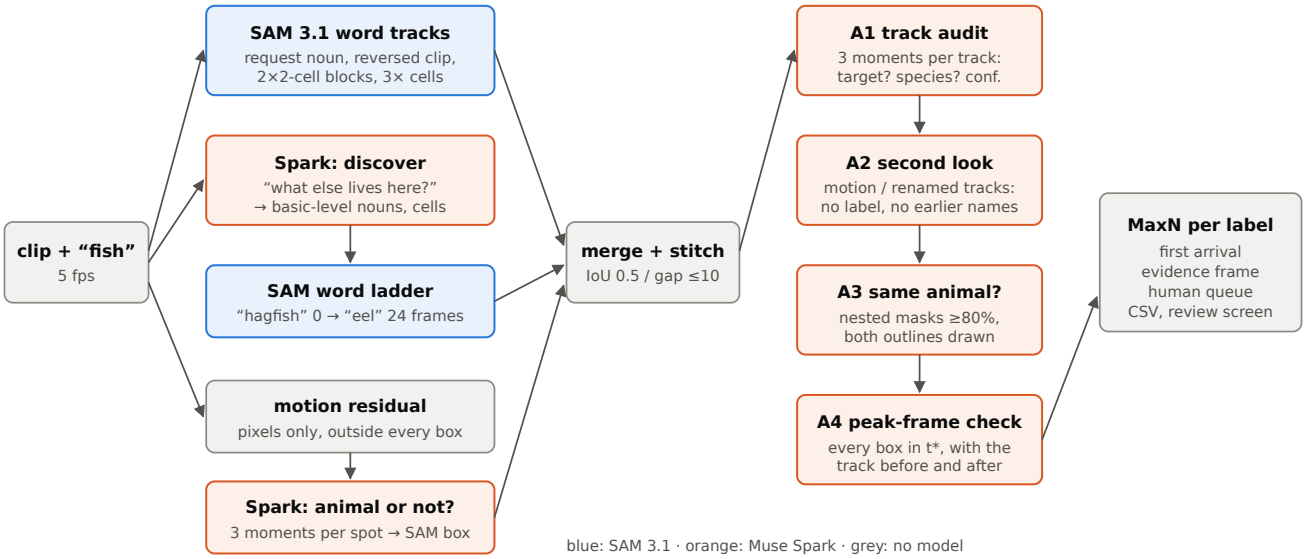


Figure 2. Pipeline. Blue boxes call SAM 3.1, orange boxes call Muse Spark, grey boxes use no model. Every Spark decision that can change a count is made on a saved sheet of crops that a reviewer can open.

4.1 Models and access

Both models were called through Meta’s hosted API: SAM 3.1 (`sam-3.1`) for video concept segmentation with text or box prompts, returning per-frame boxes, one-bit masks and confidences; Muse Spark (`muse-spark-1.3`) through an OpenAI-compatible chat endpoint with images. SAM 3.1 is billed per processed frame ($0.20 per 1,000 frames, minimum 50 frames per call); Spark per token, called with `reasoning_effort="low"` and `max_tokens=6000` (2,000 for the count-only baseline of §6.2). Neither model was fine-tuned. Every call is cached by a hash of its exact input (clip bytes and prompt for SAM; images and text for Spark), so that any run can be replayed offline at zero cost and a replay reproduces the stored decisions exactly. Because the key is the exact bytes, the clips and sheets the code builds must come out byte-identical on any machine: the x264 thread count (its output depends on it, and it defaults to the core count) and OpenCV’s Intel IPP code path are pinned. Both were found when the build machine went from 6 cores without AVX to 12 with AVX-512 and two replay tests failed; after pinning, the s002–s004 demo runs and all unit tests replay exactly on the new CPU. The brightened night clip of s001 (low light, Appendix B) still comes out different there, so a re-run of it calls the API again. Spending is capped twice: an environment variable caps the cumulative total of the project’s spend ledger, and a per-run option caps the estimated cost of one run. Spark is the cheaper of the two image-reading models the API offers (Muse Wonder costs about 3 to 5 times as much per token; no Llama vision model is offered); §7.6 puts Wonder on the same questions.

4.2 Word-prompted tracking

The clip is first tracked with each requested noun on the whole frame. Two variants address measured weaknesses (§7.4). A **reversed pass** runs the same word on the time-reversed clip and maps the detections back, because word tracking begins only once an animal is clearly visible and then runs forward. **Zoom passes** crop a region of the frame, upscale it and run the word there, for animals that are small or faint at full resolution: a 2/3-size block (2×2 of

the 3×3 cells) upscaled 1.5× around a region the VLM points to, and, if that finds nothing, the single 3×3 cell upscaled 3× with every candidate word, keeping the word that tracks longest. Detections from crops are mapped back to frame coordinates and marked "cut" when they touch the crop border.

4.3 Discovery: naming what was not asked for

Six frames of the clip are tiled into a contact sheet and Spark is asked for "every OTHER kind of living thing that the request did not ask about", with three plain-noun search words per item, an expected count, the grid cells where it appears, and indirect signs (sediment puffs, moving shadows). The prompt states the segmenter's vocabulary constraint explicitly: plain everyday nouns work; species names, colours and words like "animal" return nothing. For each proposed item (the prompt allows four; the pipeline keeps the first three), a **word ladder** tries Spark's own short name first and then its search words on the full clip, stopping at the first word that tracks; the track keeps Spark's name ("hagfish") while the word ("eel") is only SAM's handle. If the word tracks fewer animals than Spark expected, the zoom passes above run in the cells Spark named; if an indirect sign names a suspected animal, its cell is searched with that word.

4.4 Motion residual: looking where no word works

A BRUV camera is still, so the background barely changes. We compute a per-clip median frame in CIELAB at half resolution, mark pixels whose colour distance from it exceeds a robust threshold (median + 6 × MAD, at least 8) and that also changed between frames $t - 2$ and $t + 2$ by more than 0.8 times that threshold, keep blobs of 12–2,500 pixels at analysis scale, link blobs across frames into short paths, and keep paths that live at least 6 frames, have a minimum area, and lie mostly outside every box any word pass produced (boxes dilated by ±3 frames). Up to eight such spots, longest-lived and largest first, are shown to Spark as one sheet: each spot at three moments (early, at its median-size frame, late) at the same zoom. Spark answers per spot whether it is an animal, with a one-word name and a confidence; it is told that a sediment cloud changes shape across crops while an animal keeps its shape. Up to five spots it calls animals (confidence ≥ 0.6) become **box prompts** for SAM 3.1, with no word at all; the box is grown by 40%, snapped to an 8-pixel grid and kept two pixels inside the frame (a box touching the frame edge is rejected by the API). Motion alone never counts anything: it only says where to look.

4.5 Merging and stitching

All passes are merged in priority order (requested and discovered words first, then zoom, reversed and motion passes). A lower-priority detection that overlaps a kept one of the same label in the same frame is dropped, and its track is aliased to the kept track if that happens on at least half its frames and the two never co-exist. The overlap test is plain IoU ≥ 0.5 only for zoom detections that do not touch their crop border; every other detection (full-frame, reversed, motion, or a crop-cut partial) overlaps when $\max(\text{IoU}, \text{containment} - 0.2) \geq 0.5$, where containment is the intersection over the smaller box. Containment must therefore reach 0.7, and a partial animal inside a full box is recognised as the same animal. Before merging, a motion track that lies on an existing word track (IoU ≥ 0.3) in three or more frames takes that track's label and loses the overlapping frames, so one animal is not counted

under a word and a motion guess. A box drawn around two animals that other passes boxed separately is removed. Fragmented tracks of the same label are re-joined when one starts within 10 frames of the other's end, the two never co-occur, and the first box of the later one is near the last box of the earlier one.

4.6 Audits

All audit sheets share one rule learned the hard way (§7.1): **an object is always shown at three moments at the same zoom**, because across moments an animal keeps its body shape while weed sways in place and sediment spreads. Each Spark answer is a JSON list keyed by tile id; a reply that fails its shape check is never cached and is asked once more.

A1 Track audit. Every track is shown at three moments of its life, up to 12 per sheet: the frame of its largest box in the middle, and frames about a third of the track's length before and after it (shifted inward at the ends of the track so that the three crops are always three different moments). The prompt carries the detector's word and a short context string, the names the discovery step proposed for this clip (on the luderick runs, where discovery is off, this string is effectively empty). Spark returns per track `is_target`, a species name, a confidence and a reason. Confidence ≥ 0.7 with a named species sets the label; a confident rejection (`is_target` false, ≥ 0.7) removes the track; every other word-prompted or discovered track keeps the label "⟨word⟩ (unidentified)" and is counted, including an unconfident rejection. Motion-seeded tracks follow a stricter rule (§4.7). §4.7 lists which tracks are queued. For a discovered track, the question is asked about the animal Spark named ("hagfish"), not about SAM's word ("eel").

A2 Second, unprimed look. Tracks seeded from motion rest on two weak signals (it moved; one sheet looked alive), and a discovered track that A1 renames to another discovered animal rests on A1 alone. Before such a track counts, Spark sees it again at three *other* moments (1/8, 1/2 and 7/8 along its path), with no label and no names from earlier steps, and must answer "animal" with confidence ≥ 0.5; species confidence remains A1's job. This step was added after the anchoring failure in §7.5.

A3 Same animal? Two counted detections whose masks nest (one ≥ 80% inside the other) are found from the decoded masks; each pair is shown once, in the frame where it nests most, with both outlines drawn (red and cyan). Pairs of the same segmenter word and pairs of two different words (for example a renamed track and a track of another word, candidates from containment ≥ 0.6) go on separate sheets, the second kind from V6 on. Spark answers whether the cyan outline marks the same single animal or a part of it. Masks find candidates; the judgement is needed because one mask can cover two real animals (a case we observed with two overlapping nautiluses, where a zoom pass found the rear one alone). For a duplicate, the smaller-mask track is removed from every frame where it nests.

A4 Peak-frame check. For each label, every box in its MaxN frame $t^{\star}$ is shown with the same track three frames before and after, "this frame" always in the middle. Spark answers whether the middle crop shows "a whole or partial ⟨name⟩"; the prompt asks for "true" when a crop is too blurry to tell, a deliberate bias toward keeping boxes. For a label whose tracks all come from the requested word, ⟨name⟩ is that word ("fish"), not the species A1 proposed; for a discovered animal it is the animal's name ("hagfish"). Asking about the species removed real fish of another species (§7.5), which led to this rule in the final version. A rejected box re-

moves that track for ±1 s, the MaxN curve is recomputed, and the new peak is checked, for up to three rounds. This audit exists because a per-track audit can pass every track while the peak frame is wrong: in an early run on nautilus clip s004, most “shrimp” boxes in the peak frame sat on hagfish bodies although each track had passed A1.

Relabelling. A discovered or motion track that A1 confidently says is *not* its word but names another animal the discovery step had already named on its own (matched on the head noun, so that “small red crab on seabed” matches “crab”, never “seabed”) takes that name, subject to A2.

4.7 Human queue and outputs

A track is queued when it has no verdict, when its A1 confidence is below 0.7, when A1 rejected it (with any confidence, so that a person can restore a wrongly removed track), or when A2 did not confirm it. Whether a queued track counts depends on where it came from:

- **Word-prompted and discovered tracks** are removed only by a confident rejection (A1 `is_target` false, ≥ 0.7), which stays restorable from the queue. Every other queued track, *including an unconfident rejection*, is counted as “⟨word⟩ (unidentified)” until a person decides. The segmenter responded to a noun, and we prefer a visible, queued over-count to a silent miss. The cost is visible in s002 (§6.3), where a “fish” track A1 called “red crab” at 0.5 is counted as fish.
- **Motion-seeded tracks** count only when A1 accepts them with confidence ≥ 0.7 *and* A2 agrees; otherwise they are queued and not counted. Such a track rests on movement plus one look at a small crop, and in development a motion box that had drifted onto net mesh was audited as “obscured shrimp” at 0.4.

Renamed tracks counted under their new name leave the queue. A track that A1 confidently accepts as the target without naming a species (species “unknown”; 6 of 27 luderick tracks) is counted as “(unidentified)” but not queued: the queue is for decisions that change a count, and species naming of such tracks is left to the per-label review. The queue therefore mixes two kinds of item: tracks counted on weak evidence, which a reviewer should confirm, and removed tracks, which a reviewer may restore. On luderick, 7 of the 9 queued tracks are of the first kind and 2 of the second.

The outputs are per-label MaxN and first arrival, an evidence frame per label with all counted masks, an annotated video, a CSV, and a review screen that shows each queued track at three moments. Prompts are reproduced verbatim in Appendix A; thresholds are listed in Appendix B.

While the queue is open, the review screen and the CSV give MaxN as a range. The low end leaves out every queued track no person has reviewed, so only confident verdicts and human edits count; the high end is the count as reported, with queued “(unidentified)” tracks counted. Each review narrows the range and an empty queue leaves one number. Removed tracks are in neither end, so a reviewer who restores one can raise the count above the high end. In the demonstration runs the range is a single number for every label except the two “(unidentified)” rows (s002 fish 0–2, s004 shrimp 0–3).

5 Experimental setup

Counting data (luderick). Six clips from the luderick seagrass dataset (Ditria et al., 2021; PANGAEA doi:10.1594/PANGAEA.926930, CC BY 4.0; seagrass in south-east Queensland estuaries, 1920×1080), 665 frames at 5 fps, of which 319 carry human boxes (543 boxes: 527 luderick, 16 bream; we score both as "fish"). The annotations are on frames extracted at 5 fps; we re-encode each video at 5 fps so that clip frame i is annotation frame $i + 1$, and score only annotated frames. The annotations were made for training detectors on luderick, so other fish and partially visible fish are not always boxed. For this task the pipeline runs with the request "fish" and without discovery or zoom, so it measures the effect of the audits on a single-category count.

Discovery data (nautilus). The first 12 s (60 frames at 5 fps, 1920×1080) of each of the four supplementary videos S1–S4 of Barord et al. (2014) (CC BY), recorded overnight by BRUVs at 300–400 m with bait, some next to a trap. Barord et al. counted nautiluses by photo-identification, not MaxN; MaxN is our metric applied to their footage. The request is "fish" only. Before looking at any pipeline output, one of us (not a marine biologist) listed every non-fish animal visible in frames 0, 20, 40 and 59 of each clip at taxon level: 7 taxa over the four clips (nautilus in all four; a red crustacean in s002; shrimp and a hagfish in s004). This is a non-expert, taxon-level ground truth; it cannot score counts of individuals.

Held-out discovery data. Five videos from other studies, never used in development and chosen by a rule written before any run (`report/paper/discovery_heldout_protocol.md`, committed first): three baited-camera clips from a carcass study at 95 m in Saanich Inlet, British Columbia (DH1–DH3; Anderson & Bell, 2014), one from a pig-carcass study in the Salish Sea (DH4; Anderson & Bell, 2016), both CC BY 4.0, and one fish-only BRUV clip from New Zealand (DH5; Zintzen et al., 2012, CC BY 2.5). Each clip is the 12 s centred on the file's midpoint at 5 fps. DH1–DH4 were selected because their source text names a non-fish animal, so they measure finding such animals, not how often they occur; DH5 measures false discoveries. Before the run, two markers listed independently, at a coarse group level ("crab-like", "shrimp-like"), the non-fish animal shapes they could see in frames 0, 20, 40 and 59: Jiyul Ahn (not a marine biologist), by circling shapes on the sheets, and Claude, the AI assistant used to write the code, in a committed file. The bait carcasses are not animals to find. The two lists agree on every mobile animal: crab-like in DH1, DH2 and DH3, shrimp/lobster-like crustaceans in DH3, none in DH4 and DH5 (four animals-in-clip). Sessile animals (an anemone-like animal in DH2, an anemone or sea star in DH3) are scored separately. The command is the development discovery command, run once. The protocol fixes the metrics: discovery recall (reference animals that end in a counted, non-queued row of the right group), the same for names Spark proposes, false discoveries (counted non-fish rows whose evidence crop shows no animal, judged by Jiyul Ahn from the crop alone with the label hidden), queue size and cost.

Vocabulary probes. One SAM 3.1 call for "animal" on each of the 39 clips above (6 development and 24 held-out luderick, 4 nautilus, 5 held-out discovery) and one for "luderick" on the six development clips. The clip list and the reported quantities (frames with a box, most boxes in one frame, and overlap with the pipeline's counted non-fish animals) were committed in `report/paper/scripts/animal_probe.py` before the calls.

Metrics. For counting we match predicted to human boxes per annotated frame, greedily in the order of the human boxes, at IoU ≥ 0.3 (sensitivity at 0.5 and 0.7 in §7.2), and report recall (with a 95% percentile bootstrap over clips, since boxes within a track are not independent), precision, MaxN computed over the annotated frames only (for every system, so they are comparable), the number of clips with exact MaxN and with |error| ≤ 1, and the human queue as a fraction of tracks. MaxN depends only on how many boxes a system keeps per annotated frame, not on their labels or on the matching threshold. A queued track counted as "(unidentified)" is a box the system outputs, so the headline numbers assume a reviewer who accepts the queue; we also report the opposite bound, a reviewer who rejects every queued track. Replacing the greedy matching by a maximum bipartite matching changes one box at IoU 0.3 (recall 97.4% → 97.6% for SAM alone and for the audited pipeline alike) and none at 0.5 or 0.7. For discovery we report, per taxon, whether Spark named it and whether it reached the final count, and whether any label outside the person's list was counted. Queued "(unidentified)" labels are excluded from that last measure and reported separately, and a taxon is matched leniently (s002's "small red crab" and the "small blue shrimp at bait" Spark proposed both match the person's "red crustacean").

Baselines. For counting: *SAM "fish" alone*, the raw SAM 3.1 word track for "fish" on the whole frame; *SAM confidence cut-offs* (0.5–0.8) on the same track; *merge and stitch without audits*, the full pipeline with A1–A4 disabled, replayed from cache; and *merge plus a minimum track length* of k frames (k = 2–8), a model-free rule that removes short tracks. The length rule was suggested by an independent reviewer of our first draft, after the results were known; we report both its best k chosen with the test labels, which favours the baseline, and k chosen leave-one-clip-out. For discovery: *SAM "fish" alone*; *discovery without audits*, Spark's naming, the word ladder and zoom with A1–A4 disabled; and the same *plus the motion residual* (Spark's animal-or-not look on moving spots, then box prompts), still without A1–A4. Both no-audit variants are replays of the cached calls of the final runs, so they see exactly the same model replies up to the point where the audits would start. Without audits there are no species labels, so these baselines count under SAM's word ("eel", "shrimp"). Further ablations of discovery components come from the development history (§7.3).

Protocol. Results are from v16 code (luderick run tag eval6) unless marked V6 or final. V6 differs from v16 in two places: the A4 question for word-prompted tracks (§4.6) and A3 pairs across two different words (§4.6); it reproduces every v16 count exactly. Luderick run eval7 was made after the first change and before the second, which cannot act on a single-word task. The **final code, frozen at commit `cbaf59b` before the held-out re-runs**, adds to V6: A4's example of a different animal without the category word (§7.1); cross-word A3 candidates from containment 0.6; a second look or a same-group peak answer that names another animal queues the track while it still counts; animal-group MaxN rows; a separate second look for mid-confidence motion tracks; discovery names reduced to their animal noun for SAM; automatic brightening of dark clips (thresholds set from the development clips' brightness, post hoc; only s001 is that dark); and splitting of videos over 600 s. Replayed offline on the development data ($0), it reproduces every luderick number (Table 4) and every discovery count except s001, which it brightens (nautilus 2 + 1 queued → 4, none queued); s004's queue grows from 8 to 10 tracks with the same counts. Lower-case v-numbers (v13–v16) name code releases; V1–V6 name the audit variants of §7.1, of which V5 is the v16 audit. All luderick variants in §7.1 are saved runs of earlier versions of the same code, re-scored with

one scorer. Every number in this paper is produced by a script in the repository from saved runs and cached model replies (§10 says which of them a fresh clone can reproduce), except where a sentence says it is a development observation. **The six luderick and four nautilus clips are development data.** The design was iterated on them: the faint fish in frame 35 of 20190618_9 and the s004 peak frame were used as regression checks for every change, and the final layout (V5) was chosen knowing the result of V4. For counting we added, after the second review, 24 held-out luderick clips scored once with frozen code under a protocol committed before the run (§6.2), and for discovery five held-out videos from other studies, run once in the same way (§6.3). Tables 1, 2 and 3 are development numbers; Tables 1c and 3b are held-out.

6 Results

6.1 Counting against human boxes

Table 1. Luderick, six development clips, IoU ≥ 0.3. MaxN is over annotated frames; "9→10" is human MaxN → system MaxN. *Min length 4*: merge and stitch, then drop tracks shorter than four frames (no model call). *Audited*: the final pipeline (saved run eval6: v16 code, sheet layout V5; the final V6 code reproduces every count, §5), queued "(unidentified)" tracks counted (a reviewer who accepts the queue). *Queue rejected*: the same run with every queued track removed.

clip	ann. frames	SAM "fish" alone R / P / MaxN	min length 4 R / P / MaxN	audited R / P / MaxN	queue rejected R / P / MaxN	queued
20190618_9	44	98% / 72% / 9→10	98% / 72% / 9→10	98% / 72% / 9→10	75% / 72% / 9→8	4 of 14
20190709_14	126	100% / 72% / 2→2	100% / 72% / 2→2	100% / 72% / 2→2	95% / 89% / 2→2	1 of 3
20190618_3	58	100% / 98% / 2→3	100% / 100% / 2→2	100% / 100% / 2→2	100% / 100% / 2→2	1 of 3
20190710_4	40	82% / 100% / 2→1	82% / 100% / 2→1	82% / 100% / 2→1	82% / 100% / 2→1	0 of 1
20190709_13	40	100% / 100% / 2→2	100% / 100% / 2→2	100% / 100% / 2→2	100% / 100% / 2→2	0 of 3
20190709_4	11	100% / 71% / 2→3	100% / 77% / 2→2	100% / 77% / 2→2	0% / — / 2→0	3 of 3
all	**319**	**97.4% / 78.7%**, exact 2/6, Σ|err| 4	**97.4% / 79.2%**, exact 4/6, Σ|err| 2	**97.4% / 79.2%**, exact 4/6, Σ|err| 2	**84.2% / 85.3%**, exact 3/6, Σ|err| 4	**9 of 27**
constant, no model: always predict 2		exact 5/6, within ±1 5/6, Σ|err| 7	—	—	—	

A constant baseline. Human MaxN is 2 on five of the six development clips and 9 on one. Always predicting 2 (the most common value, a rule that uses no model and no video) is exact on 5 of 6 clips and within ±1 on 5 of 6, with Σ|err| 7: more exact clips than any row of Table 1, and a larger total error than the pipeline (2) because it misses the nine-fish clip by 7. MaxN-exact therefore cannot separate methods on clips whose targets are almost all 2, and the row is a reference point, not a competitor. The box recall and precision columns and the queue are the informative numbers here.

The audits remove the over-counts SAM makes, and so does a length rule. On 20190618_3 and 20190709_4 SAM alone counts 3 where people counted 2 (here and below, an over-count is a MaxN above the human one, not necessarily a false detection; see the pre-

cision caveat below). In both, A1 confidently rejected one track, and in both that track was two or three frames long and contained only unmatched boxes (recall unchanged, precision 98→100% and 71→77%). Recall stays at the segmenter's 97.4% (529 of 543 boxes): of the 529 matched boxes, the audits removed none. The boxes are not independent (they sit in 27 tracks on six clips), so we give a percentile bootstrap over clips rather than a binomial interval: 92.4–100% for SAM alone, the length rule and the audited pipeline alike, and 72.1–96.4% if the queue is rejected. With six clips these intervals are wide, and none of the counting differences in Table 1 is outside them. The 14 boxes SAM misses are of two kinds. Nine are one faint, distant fish in 20190710_4 (frames 52–61) that no SAM box touches (IoU 0), and the reason that clip's MaxN is one short; the other five are in 20190618_9, where SAM draws one box over two overlapping fish (IoU 0.11–0.31 with each human box). An audit that only removes tracks cannot bring either back. An opt-in step that shows Spark six frames with every box drawn and box-prompts SAM on what it points at (`--misslook`) recovers the faint fish here (recall 98.9%, MaxN exact), but on the 24 held-out clips it leaves recall at 98.9% and adds boxes the reference does not have (precision 77.6 → 69.4%), so it stays off by default. Merge and stitch alone change nothing on this task (identical to SAM alone), and SAM's own confidence is no help either: every cut-off from 0.5 to 0.7 leaves the output unchanged, and 0.8 drops recall to 35.7%. But dropping tracks shorter than four frames, with no model call, removes exactly the same two tracks and gives exactly the same recall, precision and MaxN (Table 1). The rule is sensitive to its threshold, which we chose with the test labels: three frames fixes only one of the two clips (exact 3/6), and five or more frames start to remove real fish (recall 95.8% at five, 94.8% at six). Choosing k fairly, leave-one-clip-out (the k that minimises the MaxN error on the other five clips), picks k = 4 five times and k = 3 once, when 20190709_4 is left out, and that clip then keeps its over-count: exact 3/6, Σ|err| 3, recall 97.4%, against 4/6 and 2 for the audits. The difference is one clip. On this single-category task, then, the audits reached the result of the best length threshold without having one tuned, and did not clearly go beyond it. We do not claim that the audits improve MaxN on luderick over a well-chosen heuristic; their distinct contributions are on discovery (§6.3, §7.3), where a length rule has nothing to say about names, duplicates of crowded animals, or a motion box on a net.

The headline recall assumes the queue is accepted. Nine of 27 tracks are queued. If a reviewer rejected all of them, recall would fall to 84.2% (exact MaxN 3/6, Σ|err| 4), and to 0% on 20190709_4, whose two real fish are both queued at confidence 0.6 and 0.65. Matching each queued track to the human boxes shows why the queue cannot simply be dropped (Table 1b): the queue holds real fish that Spark was unsure about (p0:9 in 20190618_9, confidence 0.55, matches all 44 of its boxes in annotated frames), tracks that match nothing (p0:11, 0 of 12), and the two short tracks A1 removed (no boxes in annotated frames). The size of the queue is also noisy (§7.1). Between the two cases lies the MaxN range of §4.7; it contains the human MaxN on 5 of 6 clips. The exception, 20190710_4 (1 against 2), is a fish that neither end counts.

Table 1b. Queued luderick tracks: boxes in annotated frames that match a human box, and the A1 verdict.

clip	track	length	matched / boxes	A1 verdict
20190618_9	p0:9	60	44 / 44	fish, species unknown, 0.55
20190618_9	p0:2	36	2 / 7	fish, unknown, 0.5
20190618_9	p0:14	4	2 / 4	fish, unknown, 0.5
20190618_9	p0:11	12	0 / 12	fish, unknown, 0.4
20190709_14	p0:1	56	7 / 43	fish, unknown, 0.6
20190709_4	p0:0	59	10 / 11	fish, unknown, 0.65
20190709_4	p0:1	42	7 / 11	fish, unknown, 0.6
20190618_3	p0:2	2	—	not a fish, 0.95 (removed)
20190709_4	p0:2	3	—	not a fish, 0.9 (removed)

The two MaxN errors that remain are of different kinds. On 20190618_9 (9→10), the frame that sets the count holds two real fish without a human box, one cut by the left edge; the "error" is likely in the reference. On 20190710_4 (2→1), a dark fish in murky seagrass is never detected by SAM, so no audit can add it; audits can only remove.

How long the queue takes. After the evaluation, Jiyul Ahn cleared the queue of four development clips (20190618_3, _9, 20190709_4, _14; 9 queued tracks) on a phone page that shows what the review screen shows: three crops of each track with our box and the auditor's note (`report/paper/scripts/qtime_sheet.py`, `report/paper/data/qtime/`). It took **121 s from Start to tapping Copy answers after the last answer** (the timer stops at Copy, so answering itself took at most that), about 13 s per track. Six tracks were accepted and three left unsure (an unsure track stays out of the count). Against the human boxes, every accepted track overlaps the reference on annotated frames (2 of 7, 7 of 43, 7 of 11, 44 of 44, 10 of 11 and 2 of 4 of its boxes match), and all three unsure tracks match none (one is the video's timestamp overlay, one open water, one a very faint shape). No track the reference lacks was accepted. One judge, one sitting, development clips; the judge had not seen these tracks before.

Precision is a lower bound. Unmatched boxes are not necessarily false: the annotations target luderick and omit some real fish (as in the 9→10 frame above). During development we inspected by eye 6 of 12 tracks scored as false positives in an earlier run and all 6 were unboxed fish; this is a development observation that the saved runs do not let us reproduce. A precision change from 78.7% to 79.2% should therefore not be read as the size of the audits' effect on false detections.

For context only: on footage from the same estuary, a Mask R-CNN trained on about 6,000 luderick annotations scored F1 86.8% on per-video MaxN over 31 test videos, against 85.3% for marine experts and 79.0% for citizen scientists (Ditria et al., 2020a, Table 1). The test videos, the split and the metric differ from ours and our system is not trained on luderick, so the numbers cannot be compared. They do show that a supervised single-species detector already matches experts at this site, which is one more reason we do not present luderick counting as this system's contribution.

6.2 Held-out clips

After the second review we ran the frozen final code (V6) once on every annotated video of our archive copy that development had not used: 24 clips, 6–62 frames, 445 annotated frames, 523 boxes (496 luderick, 27 bream), human MaxN 1–2. The protocol (clips by rule,

methods, metrics, one run) was committed before the run (`report/paper/heldout_protocol.md`); the length rule keeps the k = 4 chosen on the development clips. One change followed the protocol commit, 26 s later and before any held-out call: the evaluator was made to fetch SAM's "fish" reply when it was not cached (these clips had none; commit 9100f9e); the pipeline code was not touched. These clips come from the same estuary and cameras and are shorter and sparser than the development clips, so they test transfer within one site, not to new habitats.

Table 1c. Held-out luderick clips (24), IoU ≥ 0.3, frozen V6. Script: `report/paper/scripts/rescore_heldout.py`; Spark rows `spark_count_rescore.py`. Spark returns a count without boxes, so it has no recall or precision; asking it for boxes would make it a detector, which the protocol did not test. † Exploratory, not in the protocol: MaxN over the low median of five consecutive annotated frames (§6.2).

system	recall	precision	MaxN exact	within ±1	Σ|err|
SAM "fish" alone	98.9%	76.1%	19/24	21/24	14
merge + min length 4	98.9%	76.5%	19/24	21/24	13
audited, queue accepted	98.9%	77.6%	20/24	21/24	12
audited, queue rejected	95.6%	88.5%	21/24	24/24	3
Spark alone, per-frame count	—	—	8/24	16/24	33
Spark alone, 5-frame median†	—	—	15/24	22/24	17
constant, no model: always predict 1 (the most common human MaxN)	—	—	17/24	24/24	7
constant, no model: always predict 2 (the development-clip value)	—	—	7/24	24/24	17

A constant baseline. Human MaxN is 1 on 17 of the 24 held-out clips and 2 on the other 7. Always predicting 1, the most common human value, is exact on 17 clips, within ±1 on all 24, with Σ|err| 7 (Table 1c, computed from the same per-clip human MaxN as the rescoring script; the last row predicts 2, the value the development clips would suggest). The four pipeline rows are exact on 19 to 21 clips, two to four more than a rule with no model. MaxN-exact and within ±1 cannot separate methods when the targets are 1 or 2: the queue-rejected row's 24/24 within ±1 is also reached by the constant. Box recall and precision, which a constant does not have, and the queue are the informative numbers on these clips. The audited pipeline's one extra exact clip over SAM alone (20 vs 19) is read in that light.

Re-run of previously seen data on the final code. After the DH6–DH11 runs, the 24 clips were run once more with the code frozen at `cbaf59b` (tag `frz`, $0.08; log `report/paper/data/heldout/frz_rerun.log`). With the queue accepted the result is the same as the V6 run on every count: recall 98.9%, MaxN exact 20/24, within ±1 21/24, Σ|err| 12, the same

four clips off, and 15 of 49 tracks queued; precision is 77.2% against 77.6%. These clips were seen before this code was frozen, so this is not a second held-out test.

With the queue accepted, the audits fix one over-count the length rule cannot (20190618_6, 1→2 becomes 1→1: A1 rejected a 6-frame track as "empty blue water, no fish" at 0.85, too long for k = 4) and otherwise leave the segmenter's output as it was: three clips keep large over-counts (20190709_17 2→7, 20190709_9 1→5, 20190709_11 2→4). In the first two most tracks are queued (5 of 8, 4 of 5); in 20190709_11 only 2 of 6 are. The queue holds 15 of 49 tracks. Rejecting it brings every clip within ±1 (Σ|err| 3) at a recall cost of 3.3 points overall, but the cost falls on real fish: recall drops from 100% to 52% on 20190709_15 and from 79% to 50% on 20190710_9, whose MaxN becomes an under-count (2→1). We looked at the peak frame of the worst clip by eye: all seven boxes in 20190709_17 frame 15 are fish-shaped, three clearly fish and four faint ones, against two human boxes, so much of the "over-count" is likely unboxed fish rather than false detections (consistent with the precision caveat of §6.1).

Checking the outliers: the reference, not the model. The three clips with the largest over-counts (20190709_17, _11 and _9) hold 145 of the 153 held-out boxes scored as false. After the runs, Jiyul Ahn counted their MaxN by hand from the full-rate videos, without boxes: 8, 3 and 5 fish. The human boxes give 2, 2 and 1; the pipeline (queue accepted, final code) gives 7, 4 and 5. Against this count the pipeline is off by 2 in total and the boxes by 11. Figure 3 shows the last frame of 20190709_17: every pipeline box is on a fish, the human boxes cover two of them, and one fish in the seagrass is boxed by neither, which is the eighth fish of the hand count. The luderick annotations were made to train a single-species detector, not to count every fish, so on these clips a "false positive" is mostly an unboxed fish. This is one person and three clips, chosen because they were outliers; it explains the low precision on them but is not a corrected precision for the set.

A corrected precision. To measure one, 40 of the 153 unmatched held-out boxes were drawn at random (seed and scoring rule fixed before judging) and mixed with 10 matched boxes; Jiyul Ahn judged each crop with labels hidden as fish, not fish or unsure (`report/paper/data/heldout/fp_sheet/`). All 10 matched boxes were called fish. Of the 40 unmatched boxes, 29 were fish, 2 were boxes on part of a fish, 9 were unsure (7 of them on tracks the pipeline had already sent to the person queue). **None of the 40 was called not a fish.** Held-out precision is therefore 93.7% counting only the 29 clear fish as correct (95% Wilson interval 90.2–96.3%), 94.9% with the two part-of-a-fish boxes, and 100% of the sample if the unsure boxes are the fish they were drawn on, against the 77.2% scored on the reference boxes. One judge, the first author, and one sample of 40.

A second reference for all 24 clips. Prompted by the three outliers, Jiyul Ahn then counted MaxN by hand on the other 21 held-out clips, from the full-rate videos without boxes, in a shuffled order under neutral file names (key fixed before counting; `report/paper/data/heldout/recount/`). Where he saw fish he could not tell from moving seagrass, he gave them separately (two clips, +1 and +2). Against this count (sure fish only):

Table 1d. Held-out MaxN against a hand count of the full videos (24 clips).

system	MaxN exact	within ±1	Σ|err|
human reference boxes (luderick annotation)	16/24	22/24	16
pipeline, queue accepted, annotated frames (as Table 1c)	18/24	24/24	6
pipeline, queue accepted, every frame of the clip	17/24	24/24	7

With the faint fish added to the hand count, the pipeline (annotated frames) is exact on 17/24 with Σ|err| 9 and the boxes on 15/24 with 19. The pipeline's remaining errors are four under-counts by one (20190618_6, 20190710_1, _3, _8), where the boxes miss the same second fish, and one over-count by one (20190709_11). Against the boxes it was judged on in Table 1c, the same output scores 20/24 and Σ|err| 12; against an independent count of the videos it is closer than the boxes are. The counter is the first author and had seen the three outlier clips flagged before he counted them; the other 21 were counted blind to clip identity and to every count, though some of their pipeline boxes had appeared as unlabelled crops in the precision sheet above.

Correcting the reference in ten minutes. After the evaluation we turned this check into a tool (`fishcount/gapfind.py`, README): it collects every counted track whose boxes match no reference box on annotated frames and asks one question per track on a phone page, showing three crops with our box in yellow and the existing reference boxes in grey. On the held-out set the 153 unmatched boxes come from 20 tracks. With three matched control tracks mixed in, Jiyul Ahn answered the 23 cards in 9 min 39 s (first to last tap). All three controls were answered "already boxed". Of the 20 tracks, 15 were unboxed fish (145 boxes added), one was a fish the reference had already boxed differently, one a box on part of a fish and three unsure; **none was "not a fish"**. The corrected annotation raises MaxN on four clips (20190709_9 1→5, _11 2→4, _15 2→3, _17 2→7) and matches the hand count of Table 1d as closely as the pipeline does (exact 18/24, within ±1 24/24, Σ|err| 6, against 16/24, 22/24 and 16 for the original boxes). The added boxes are the pipeline's own, so we do not re-score the pipeline against this annotation; what it measures is the cost of the fix: under ten minutes of taps in place of drawing 145 boxes. The judge had seen 40 of the 153 boxes in the precision sheet. That overlap doubles as a test–retest check of the judge. Fifteen tracks appear in both sheets, judged first as single boxes and then as tracks of three crops. All 11 tracks with at least one box called a fish in the first sheet were called fish again, and no answer turned into "not a fish" (none was given in either sheet). Of the four tracks answered only "unsure" the first time, two were called fish when seen as a track and two stayed unsure.

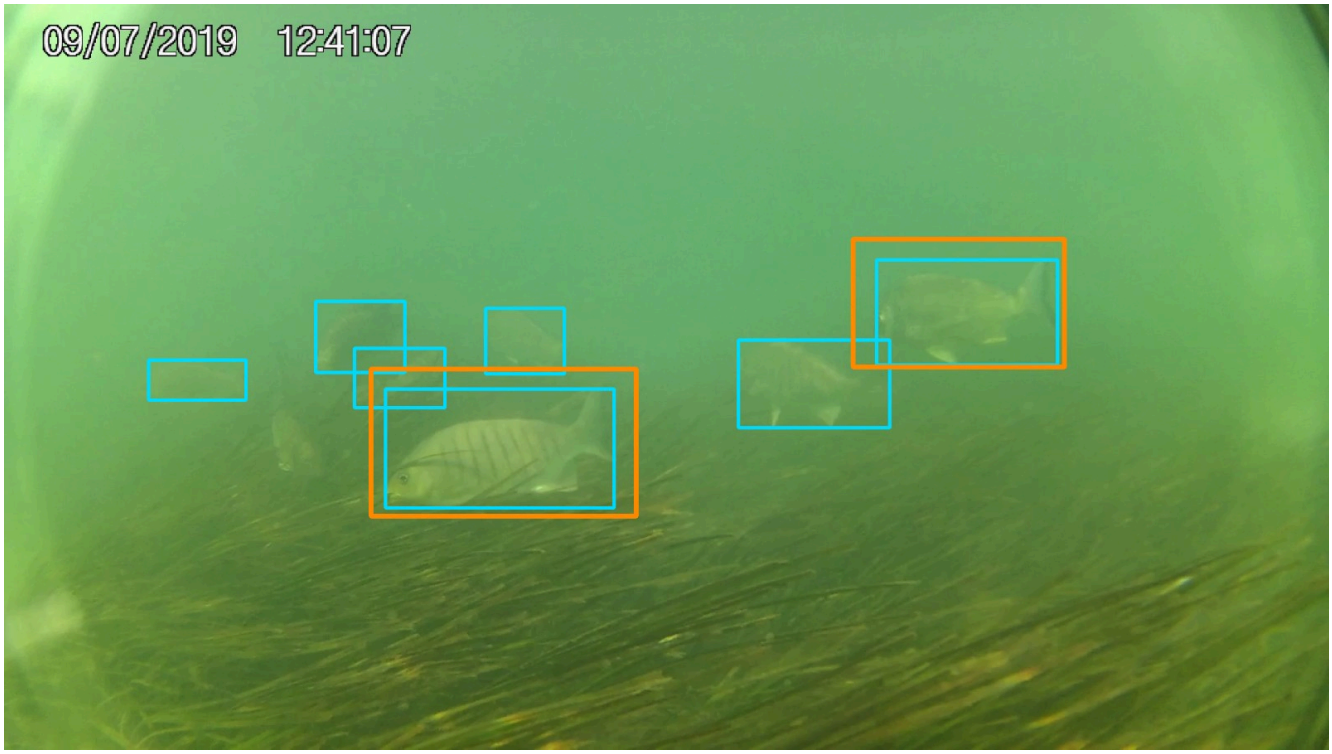


Figure 3. The last frame of held-out clip 20190709_17. Cyan: pipeline boxes (7). Orange: the human reference boxes (2). The pipeline's five "false positives" here are fish; the fish in the seagrass at left is boxed by neither. A hand count of the clip gives MaxN 8.

The held-out result therefore repeats the development finding in a weaker form: the audits do not beat the segmenter or a length rule on box metrics, and their useful output on this task is the queue, which contains most of the tracks behind the clip-level over-counts, but also real fish Spark was unsure of; it narrows a person's work rather than replacing it. Read as the range of §4.7, from queue rejected to queue accepted, the held-out MaxN is a single number on 19 of 24 clips and contains the human MaxN on 22 (mean width 0.46). Both misses lie above the human count: 20190709_11 (3–4 against 2) and 20190709_17 (3–7 against 2), the clip whose peak frame shows unboxed fish. The widest ranges (3–7, 1–5) are on the two clips with the largest over-counts, so the width points a reviewer to them. The range was defined after these results were seen; it summarises the two protocol rows above and is not a protocol measure.

The VLM alone gives counts that cannot be checked. The second review asked whether Spark could replace the segmenter. Under a protocol committed before any call (`report/paper/spark_count_protocol.md`), Spark counted the fish in every annotated frame of both sets, one call per frame with the frame alone, and its MaxN is the largest of those counts. It ran with the audits' model and reasoning effort, but with the frame downscaled to 1280 px (SAM sees 1920 px), `max_tokens=2000` (audits: 6000) and one prompt, which asks for every fish "including fish that are partly visible, cut by the frame edge, or faint" (`report/paper/scripts/spark_count.py`). On the six development clips its MaxN is within ±1 everywhere but exact on only 1 of 6 (Σ|err| 5, all above; SAM alone: 2 of 6, Σ|err| 4). On the held-out clips it is exact on 8 of 24, within ±1 on 16 and Σ|err| 33, never below the human count (Table 1c). Per frame the picture is closer: Spark's count equals the number of human boxes

in 217 of 319 development frames (68%) and 311 of 445 held-out frames (70%), is above it in 94 and 133 and below it in 8 and 1; the per-frame mean absolute error, the protocol's secondary measure, is 0.40 and 0.53. Part of the clip-level gap is the statistic itself: the maximum of many independent noisy counts is biased upward, while a tracker's counts are coherent in time. As an exploratory check that was not in the protocol, MaxN over the low median of five consecutive annotated frames is exact on 4 of 6 and 15 of 24 clips (held-out within ±1 22 of 24, Σ|err| 17; `report/paper/scripts/spark_count_rescore.py`, which re-scores everything here from the bundled replies), still behind SAM alone on the held-out set.

Whether the remaining excess is false counting cannot be decided against these boxes. Spark's largest excesses fall on the clips where SAM also exceeds the boxes (20190709_17: Spark 8, SAM 8, human 2; 20190709_9: 6, 5, 1), where we have already seen unboxed fish. On 11 of the 19 held-out clips where SAM's MaxN equals the human count, Spark's is higher, by up to 3. We looked by eye at the frame where Spark's count most exceeds the boxes on four of them (Figure 4): each shows, besides the boxed fish, two or three faint fish-like shapes without a box. On such clips the agreement of SAM and the boxes may be agreement in leaving faint fish out, and the prompt asked Spark to count exactly those fish. Because Spark's reply is a number without locations, we cannot tell whether those shapes are what it counted; a person cannot check the number, and nothing links it across frames into a MaxN with per-number evidence. That, more than its error against these boxes, is why the pipeline keeps the division of labour: the segmenter boxes, and the VLM names and judges boxes it is shown. One call returned an empty reply (all 2,000 tokens spent on reasoning). The script committed with the protocol stopped on such a reply; we changed it to make the one further call the protocol allows (and to send calls in parallel) and re-ran it with every earlier reply served from the cache (commit eeb952b). No frame is missing. The 765 calls cost $2.47.

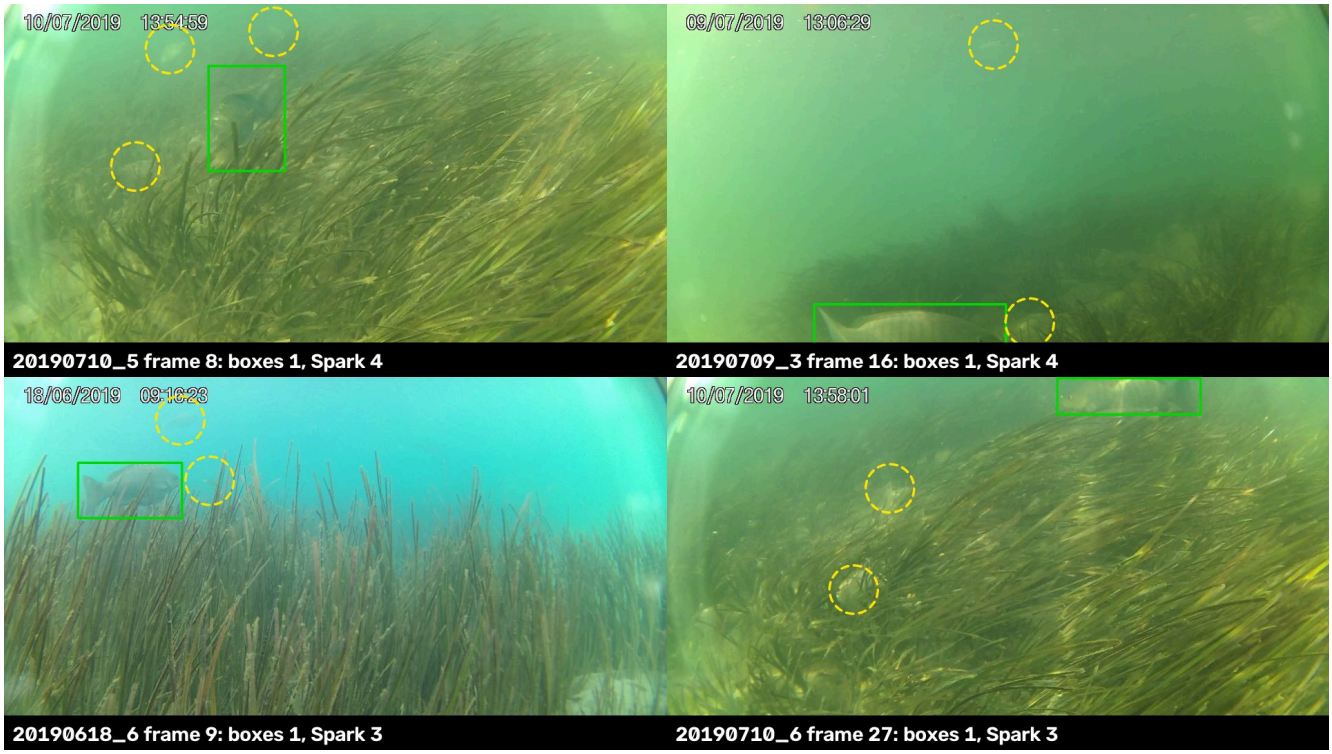


Figure 4. The frame where Spark's count (Spark alone) most exceeds the human boxes, on four held-out clips where SAM's MaxN equals the human count. Green: dataset boxes. Dashed yellow: fish-like shapes without a box, marked by us from eye inspection (non-expert). Spark's reply has no locations, so these are not known to be what it counted. Script `report/paper/scripts/fig_spark_over.py`.

Is the auditor's confidence informative? Pooling the 74 luderick tracks A1 judged that have boxes in annotated frames (27 development, 47 held-out), the share of tracks whose boxes mostly match a human box rises with A1's confidence: 0 of 7 below 0.5, 5 of 11 (45%) at 0.5–0.7, 20 of 26 (77%) at 0.7–0.9 and 25 of 26 (96%) at 0.9 or above (Brier score 0.13). The 0.7 threshold therefore sends the band where real and spurious tracks mix to a person: the queue holds 22 tracks, of which 6 match, against 45 of 52 outside it. Because the boxes omit some real fish, these shares are lower bounds. Script: `report/paper/scripts/calibration.py`.

6.3 Finding what was not asked for

Table 2. Nautilus development clips, request “fish”. Person = non-expert, taxon-level list from four frames per clip, made before looking at outputs.

clip	person listed (not fish)	final count (MaxN)	how it was found
s001 (night, wire trap)	nautilus: 1 in trap, 3+ outside	nautilus 4, none queued (final code, clip brightened); v16 on the original clip: 2 (+1 queued)	motion residual; on the original clip every word 0 frames
s002	nautilus 1; red crustaceans 2	nautilus 1; small red crab 1 (+2 “fish (unidentified)” queued)	“nautilus”; crab via “shrimp” in a 3× cell, renamed by A1
s003	nautilus 3–4	nautilus 3	“nautilus” forward, reversed and zoomed
s004	nautilus 1; shrimp 10+; hagfish 1	nautilus 1; shrimp 15 (A1: “caridean shrimp” 12, “unidentified” 3); hagfish 1	hagfish via “eel”, renamed; “shrimp”; 2 shrimp via motion

How each taxon was reached, from the run logs. **s001**: “nautilus”, “shell” and “snail” track 0 frames on the full frame, two blocks and two 3× cells of this night clip; the motion residual finds moving spots that Spark calls “nautilus”, and box prompts on them track 56 and 60 of 60 frames (A1 0.95 and 0.9, A2 yes). A third motion track (A1 0.55) is queued (v16). The final code first brightens this clip (median Lab L 5, mean 26.5; the thresholds, median < 15 and mean < 35, were set from the development clips’ brightness, so the rule is post hoc, `report/paper/lowlight_s001.md`): “shell” then tracks one nautilus for 9 frames, four motion box prompts track 50–60 of 60 frames, all four pass A1 and A2, and the peak check keeps 4 of 4 — nautilus 4, none queued. **s002**: “nautilus” tracks 48 frames. “crab” tracks 0 frames on the full frame; in the 3× cell “crab” tracks 5 frames and “shrimp” 12, and A1 calls the “shrimp” track “small red crab, not shrimp” (0.7), which matches Spark’s discovery name, so the track is relabelled; A2 agrees. **s003**: “nautilus” tracks 33 frames forward, 60 reversed and 45 and 44 in two zoom blocks. Spark also proposed a “large striped shrimp” that no word tracked; a motion box audited under that name was rejected by A2 (§7.5). **s004**: “hagfish” tracks 0 frames and “eel” 24, counted as hagfish; “shrimp” tracks 40 frames forward and 60 reversed; two shrimp at the bottom edge are reached only through motion box prompts (30 and 55 frames after merging).

Table 3. Discovery summary over the four development clips (7 taxa). The no-audit baselines replay the final runs' cached calls with A1–A4 disabled; a track counts under SAM's word, or, for a motion box prompt (which has no word), under the one-word name Spark gave the moving spot.

measure	SAM "fish" alone	discovery, no audits	+ motion, no audits	audited pipeline
taxa in the final count	0 of 7	6 of 7	7 of 7	**7 of 7**
… of which under a wrong name	—	2 ("eel" for hagfish, "shrimp" for crab)	2	**0**
non-fish labels counted that match nothing on the person's list	0	0	1 (s003 "shrimp" 2)	**0**
s001 nautilus (person: 4 or more)	0	0	3	2 (+1 queued); final code, brightened: 4
s003 nautilus (person: 3–4)	0	3	4	3
s004 shrimp (person: 10 or more)	0	15	18	15 (12 + 3 unidentified, one frame)
"fish" counted (the person listed non-fish animals only)	s002: 2	s002: 2	s002: 2; s004: 1 (motion, "small fish swimming away")	s002: 2, both queued; s004: 0

The naming step finds the taxa; the audits fix names and counts. Spark proposed nine items over the four clips. Eight correspond to animals in the person's list; one, s003's "large striped shrimp", corresponds to nothing the person found. SAM prompted with "fish" found none of the seven taxa (in s002 it boxed two objects as low-confidence "fish", at least one of them a crustacean; see below). Spark's naming plus SAM's word ladder and zoom, with no audit at all, already counts six of the seven taxa, and adding the motion residual (Figure 5), still without audits, reaches seven of seven (Table 3). The taxon score of the full pipeline is therefore owed to discovery, not to A1–A4. What the audits change is visible in the other rows. Without them the hagfish is counted as "eel" and the crab as "shrimp", because SAM's handle is the only label; with them both carry the animal Spark named. With the motion residual but no audits, the s003 trap mesh is counted as two "shrimp" (a false taxon), and A2 removes it. In s004 the shrimp MaxN, counted over both shrimp labels in the same frame, drops from 18 to 15: A3 found six nested pairs that were one shrimp counted twice, and A4 removed a box on the trap's pipe; these act in different frames, so the peak falls by three, not seven. A1 also splits those 15 into "caridean shrimp" (MaxN 12) and "shrimp (unidentified)" (3), which peak together in frame 54; a per-label output shows 12 (the final code adds an animal-group row, "crustacean 15", to the result, CSV and review screen), but that drop is the label fragmentation of §7.5, not a correction. The person's "10 or more" from four frames cannot say whether 15 is right, so we report the change without scoring it. The audits also cost one animal: in s001, the third motion nautilus has A1 confidence 0.55 and waits in the queue, so the audited v16 count (2) is further from the person's "4 or more" than the unaudited one (3). On the brightened clip (final code) all four motion nautiluses pass the audits (4, none queued).

A fixed noun list instead of the VLM's names. The second review asked whether a list of common nouns would do as well as Spark's naming at no VLM cost. We prompted SAM 3.1 on

the full frame of each clip with 16 basic-level nouns for animals seen on baited cameras (fish, shark, ray, eel, crab, shrimp, lobster, octopus, squid, snail, shell, starfish, jellyfish, sea urchin, sea cucumber, worm), written before the calls and without “nautilus”, and again with “nautilus” added as the survey’s known target ($0.55, 46 new calls; `report/paper/scripts/fixed_vocab.py`). A word track counts as finding a taxon when it overlaps a track the audited run counted under that taxon (IoU ≥ 0.3 in at least three frames). The generic list finds 5 of the 7 taxa, but only one (s004 shrimp) under a right word: the s002 crab is found as “octopus”, the nautiluses as “snail” and “shell”, the hagfish as “eel”, and the s003 nautilus not at all. Adding “nautilus” reaches 6 of 7, as many as Spark’s naming without audits (Table 3), with four taxa under a right word. The comparison favours the pipeline in two ways: the fixed list ran on the full frame only, while the pipeline’s words also get reversed passes and zoom passes (1.5× block, 3× cell) (the s002 crab is reached by the pipeline only in a 3× cell), and a taxon counts as found only when a list track overlaps a track the audited run counted, so the list can never be credited with an animal the pipeline missed. On these clips a fixed list is thus at least as good at finding taxa as Spark’s naming without audits. On s001 “octopus” tracks a spot for 57 frames that matches no counted animal; the baseline cannot say whether it is the uncounted nautilus inside the trap or nothing. That run used the original dark clip; on the brightened copy that the final code tracks, the 16 words plus “nautilus” box one of the four counted nautiluses, as “shell”, in 9 of 60 frames, and the other 16 words return nothing ($0.17, `report/paper/data/fixed_vocab_brightened/`). The final run counts all four: Spark’s word “shell” gives the same 9-frame track, and each of the four animals is also tracked from a box with no word on a moving spot Spark picked (50, 60, 59 and 60 of 60 frames; frames already tracked are not counted twice; side-by-side video `report/scripts/split_s001_night.py`). A fixed list thus recovers most taxa on these clips if it happens to contain the target, but it names them by whichever word fired; what the VLM adds is the name, and a list it did not need to be given.

The two queued “fish” in s002. The audited output for s002 also lists “fish (unidentified): 2”, from two low-confidence “fish” tracks that sit in the queue; the person listed no fish. One (A1: “red crab”, 0.5) co-exists with the counted crab for 42 frames without overlapping it, so it is probably the second red crustacean the person saw, missed as a crab and left under the requested word. The other (A1: “dark elongated shape”, 0.4, at the top edge) is unidentified. Both are counted until a reviewer decides (§4.7: a word track is removed only by a confident rejection, and the first was rejected at 0.5), so the s002 output overstates “fish” by at least one unless the queue is worked. The person’s list does not cover fish, so Table 3 reports “fish” counts separately instead of scoring them; the s004 “fish” of the motion-only baseline is a moving spot that A1 rejects in the audited run (0.6, “box tracks shadow”).

Counts of individuals are not scored: four frames per clip are not enough for a person to count ten or more moving shrimp. In s001 v16 counted the taxon with a number (2) below what a person sees (4 or more); the final code, which brightens this clip, counts 4.

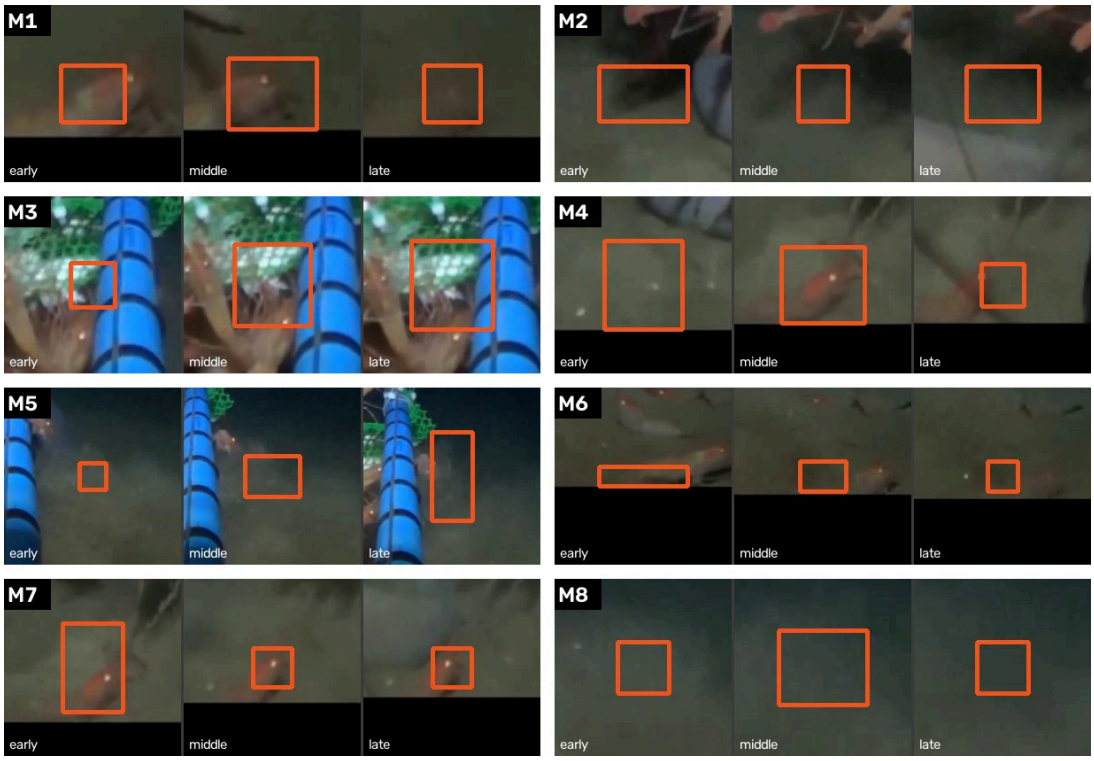


Figure 5. Motion-residual sheet, s004. Each group is one moving spot no box covered, shown early / middle / late at the same zoom. Spark calls the shrimp animals and the sediment puffs not; the shrimp then go to SAM 3.1 as box prompts with no word.

Held-out clips. Table 3b is the one run on the five held-out videos (§5), with the request "fish" and the code frozen at the protocol commit.

Table 3b. Held-out discovery, one run. Reference = the non-fish animals both markers listed before the run (sessile animals in brackets, scored separately). Queue = tracks sent to a person.

clip	reference (not fish)	Spark named	counted (MaxN)	queue	cost
DH1 (carcass, 95 m)	crab-like	crab	crab 1	1	$0.10
DH2 (carcass, 95 m)	crab-like [anemone-like]	sea anemone, crab, eel	crab 1; [sea anemone 1]	5	$0.30
DH3 (carcass, 95 m)	crab-like; shrimp/lobster-like [radial animal]	spider crab, squat lobster shrimp, sunflower sea star	spider crab 3; squat lobster shrimp sent to the queue	6	$0.40
DH4 (pig carcass)	none	—	none	0	$0.02
DH5 (fish-only BRUV)	none	small shrimp	no non-fish animal	14	$0.24
total	**4 animals**	**4 of 4**	**3 of 4 counted, 1 queued; 0 false**	26	$1.06

Spark names all four reference animals and the pipeline counts three. The fourth, DH3's shrimp-like crustaceans, is reached the way s004's bottom-edge shrimp were: "shrimp", "prawn" and "lobster" track 0 frames, the motion residual finds an "upright orange shrimp",

and a box prompt tracks it in all 60 frames. A1 names it “squat lobster shrimp” at confidence 0.5, below the 0.7 a motion track needs to count, so it goes to the person queue with its name and evidence instead of into the count (§4.7). Every counted non-fish row shows an animal: shown the evidence crops with the labels hidden (`report/paper/data/discovery_heldout/crops_sheet.jpg`), Jiyul Ahn called all four of them animals; DH2’s crab box also covers the carcass it sits on, which does not change the count. On the two clips without a non-fish animal nothing non-fish was counted. DH4 produced no track. In DH5 the counted rows are “fish (unidentified)” (MaxN 8) and “pink cusk-eel” (1), a fish whose name matches the species caption burned into that video; Spark’s “small shrimp” there, like its “eel” in DH2 and s003’s “large striped shrimp”, was tracked by no word and counted nowhere. Of the sessile animals, DH2’s is counted as “sea anemone”; DH3’s is named “sunflower sea star”, and “starfish”, “star” and “urchin” track 0 frames. The per-clip rows, evidence crops and judgement are in `report/paper/data/discovery_heldout/`.

A second run. After scoring, we ran the same command once more on the five clips with an empty cache (`runs/DH*_rerun`, 112 calls, $1.08), so every model call was fresh. It counted the same three crab-like animals and DH2’s anemone, again counted no false discovery, and again sent DH3’s shrimp-like animal to the queue (found by the motion residual as a “vertical red shrimp”, tracked in 60 of 60 frames, A1 confidence 0.5). Spark’s proposed names differed in wording (DH3: “crab”, “starfish”, “eel”; DH5: “tiny swimming crustaceans”, tracked by no word), and the queue held 27 tracks against 26.

Re-run of previously seen data on the final code. The five clips were run once more with the code frozen at `cbaf59b` (`runs/DH*_frz`, $0.03, mostly cached; `report/paper/data/discovery_heldout/frz_rerun.md`). The counted rows are the same as in Table 3b (crab 1; crab 1 and sea anemone 1; spider crab 3; nothing; fish only) and DH3’s shrimp-like animal again stays out of the count. New with the final code are three non-fish rows that are in the MaxN but queued (DH2 “coral (unidentified)” 0.4 and “crab (unidentified)” 0.45, DH3 “crab (unidentified)” 0.6): a person sees them as counts with a review flag. They were not judged with labels hidden and are reported apart.

A second held-out set (DH6–DH11). Under a second protocol (`report/paper/discovery_heldout2_protocol.md`, amendment 2: code frozen at `cbaf59b`), six more videos from other PLOS ONE studies were run once: two fish-only shark videos (a BRUV and a cleaning-station camera), a cold-seep time-lapse (two images an hour), two sediment-profile “Wormcam” time-lapses and a hydrothermal-vent observatory clip. Before any run, Jiyul Ahn and a second person (an undergraduate who had not seen any output of the project and marked alone) listed the non-fish animals on the same label-free four-frame sheets; Claude’s list is reported for agreement only. Both people listed one animal in the six clips, a bug- or worm-like animal in DH11, and nothing in DH6–DH10.

Table 3c. Second held-out discovery set, one run with code frozen at `cbaf59b`. Reference = non-fish animals any person listed before the run. Baseline = SAM 3.1 with the 16 fixed words of §6.3.

clip	reference (not fish)	Spark named	counted (MaxN)	queue	cost	baseline non-fish words with tracks
DH6 (BRUV, tiger sharks)	none	giant clam beside bait	shark 1, leopard shark 1 (fish)	0	$0.21	—
DH7 (cold seep, time-lapse)	none	crab	**crab 3**	2	$0.09	crab (10 tracks), starfish (2, on the crab)
DH8 (cleaning station, threshers)	none	thresher shark	fish 15, thresher shark 2 (fish)	5	$0.10	—
DH9 (Wormcam)	none	—	none	1	$0.04	—
DH10 (Wormcam)	none	pale branching worm	none	0	$0.18	—
DH11 (vent, 2186 m)	bug/worm-like	scavenging amphipod crustacean, colonial tube worms	amphipod 1	0	$0.23	worm (1, on the amphipod)
total	**1 animal**	**1 of 1**	**1 of 1 counted; 0 false**	8	$0.86	baseline $1.00

The pipeline counts the one reference animal, as an amphipod (the scoring note fixed before the run counts a worm or a bug-like small crustacean for “bug/worm-like”), and the fixed-vocabulary baseline finds it too, as “worm”. Shown the two counted non-fish crops with labels hidden, Jiyul Ahn called them “a crab” and “a bug/worm or a crustacean”: no false discovery. On the two fish-only clips neither method produced a non-fish row. Spark’s other proposals (a giant clam in DH6, a branching worm in DH10, tube worms in DH11) ended in no counted row.

The crabs of DH7 are the one result outside the protocol’s metrics. Neither person listed an animal in DH7, yet the pipeline counted three crabs, at their peak in frame 34; none of its crab boxes falls on the four frames (0, 20, 40, 59) of the markers’ sheet, since the time-lapse shows each crab in only a few images (the counted tracks lie in frames 14–58 and cover 1 to 11 frames each). The label-free judge called the evidence crop “a crab”, the source paper’s description of the video names lithodid crabs, and the baseline’s “crab” tracks overlap the counted one. They are not in the recall, since the reference list did not have them, and not a false discovery. We read it as a limit of four-frame reference lists, not as the pipeline seeing better than a person who watches the whole video. The fixed-vocabulary baseline called the same crab “starfish” as well as “crab”, and made one “shark” track on this clip where no person saw a fish. With a single reference animal this set mainly tests false discoveries, of which there were none. Rows, crops, judgements and the markers’ sheets are in `report/paper/data/discovery_heldout2/` (`results.md`).

How these six clips were chosen. By the protocol's rule, a clip was taken when its file description or source article names a non-fish animal, or, for the fish-only clips, names fish only (`report/paper/discovery_heldout2_protocol.md`; every candidate is logged in `report/paper/data/discovery_heldout2/candidates.md`). Four clips (DH7, DH9, DH10, DH11) were taken because their source text names a non-fish animal, and two (DH6, DH8) as fish-only. The pipeline counted a non-fish animal on two of the four (DH7, DH11) and none on DH9 and DH10. This is not a recall on the clips: the source text describes whole videos, while each clip is a 12-s window that need not show the animal. The fixed 16-word baseline also found the DH11 animal (as "worm") and the DH7 crab (as "crab", and as "starfish" on the same crab); what the VLM adds on these two clips is the name, "amphipod" and "crab", not the finding.

6.4 Cost and time

All model calls made during development and evaluation, including discarded experiments, cost $3.04 (450 calls: 148 SAM, 302 Spark; mean Spark call $0.0044, median 653 input and 684 output tokens). The repeated-call experiment of §7.1 added 98 Spark calls ($0.54, including the 25 calls of the example test in §7.5 and one call lost to an interrupted script), the held-out runs 80 calls ($0.42) and the fixed-vocabulary baseline of §6.3 46 SAM calls ($0.55), the Spark-alone counting baseline 765 Spark calls ($2.47), the vocabulary probes of §7.4 45 SAM calls ($0.62), the held-out discovery runs 106 calls ($1.06) and their uncached second run 112 calls ($1.08), later development of the final code (low light, long videos, fixes) 153 calls ($1.82), the second held-out discovery set DH6–DH11 87 calls ($0.86) and its fixed-vocabulary baseline 83 SAM calls ($1.00), and the re-runs of previously seen data on the frozen code 42 calls ($0.11), empty-cache live rehearsals of DH11 and s003 (two takes each) 48 and 40 calls ($0.51, $0.41), site-memory demo runs 3 calls ($0.02), the fixed list on the brightened clip of §6.3 14 SAM calls ($0.17), the Muse Wonder comparison of §7.6 101 calls ($2.33), the SAM 3 Agent-style baseline of §7.7 44 calls ($0.41), the 53-min deployment below 82 calls ($17.27 over three attempts), and, on the final code, development after the freeze with the re-runs of every frozen evaluation 601 calls ($4.33) and the re-runs of the 53-min video 269 calls ($5.67), for a ledger total of $44.68 over 3,269 calls; `report/paper/scripts/ledger_summary.py` produces every number in this paragraph from the spend ledger (a copy is in the scoring bundle). From the call counts of the final runs, a fresh discovery run on a 12-s clip makes 21–30 calls and costs an estimated $0.22–0.27, dominated by SAM passes (15–19 per clip, each billed for the clip's 60 frames, $0.012 per pass). The spend ledger does not separate the luderick counting runs from development calls; by the price list, one SAM pass over their 665 frames costs $0.13, and the audits add a few Spark calls per clip. A cached replay of a discovery clip, including rendering the annotated video, takes 13–24 s on a CPU server; live runs took up to 98 s, bounded by API latency. These are measurements on clips of 12 to 53 s; SAM 3.1 accepts at most 10 minutes per file, so the final code cuts a longer video into equal back-to-back segments of at most 600 s and merges them into one ordinary run (`fishcount/longvideo.py`, `report/long_video.md`). Our longest source is 40 s (Barord et al. S3): in one pass it counts the same 3 nautiluses as its 12-s clip ($0.35, 297 s live; replayed on the frozen code: 3). Forced 20-s segments counted 1, for reasons inside a segment (one SAM ID covering two animals; a five-word name never tried as a word, which the final code's short names address, not re-measured).

One full deployment has been run end to end: a 53.2-min baited reef-fish recording from NOAA's Southeast Fisheries Science Center (Data statement), 15,945 frames at 5 fps in six 600-s segments, with the discovery settings of Table 2, no code change for this video and no ground truth, so it measures cost, time and robustness, not accuracy (`report/paper/data/long_video/README.md`). The completed attempt cost $15.08 (68 live calls) and took 3.5 h run sequentially; segments 2–6 cost $1.63–5.88 and took 23–57 min each (2.6–6.4 times real time), and segment 1 reused 11 answers paid in the first attempt. The two earlier attempts ($2.18) were stopped by memory, not by the counting: drawing the annotated video held every frame of a 600-s segment in memory, and reversing the clip buffered every frame. Both now stream or work in chunks; on the development clips the outputs are unchanged (annotated video identical frame by frame; reversed frames equal up to re-encoding noise) and the unit tests pass. Per hour of video that is about $17 for the completed attempt and $19.5 counting all three, below our earlier extrapolation from the 40-s clip (about $30 and 7.5 h). The merged run reports fish (unidentified) MaxN 8 at 546 s and jack 1 at 2,133 s; in the first two minutes, while the camera is still on the boat deck, it also counts the crew (human 3, and one figure behind a trap named "person (unidentified)", a name split of the same kind as §7.5) and a fish lying in a trap on deck. The auditor rejected 35 of the 116 tracks. On the final code (clean and refill, on by default after the freeze), the same six segment clips were run again with every cached answer reused (run `LONG1_cr2`, log `report/paper/data/long_video/run_cr2c.log`; $5.67 together with an earlier partial re-run). Refill sends each segment again after a SAM call has used its 16 object IDs, so much more of the hour is tracked: 1,373 tracks after cutting reused IDs, 185 rejected by the auditor. The merged run reports fish (unidentified) MaxN 14 at 418 s, grunt 2, flatfish 1, needlefish 1 and sand lance 1 (names are proposals, §7.5); the deck crew is unchanged (human 3, person (unidentified) 1) and "jack" no longer appears. Without ground truth this shows what refill adds on a long video, not that 14 is the right count. Segments run one after another; running them in parallel is not implemented.

7 Analysis

7.1 Ablation: the layout of the evidence changes what the audit sees

We re-scored the saved luderick runs of six audit variants (Table 4). V1–V4 differ only in the audit sheets; V5 restores the V2 layout and adds steps that are inactive on this task, and V6 changes only A4's question, so V5 and V6 share one sheet design and the same counts. Figure 6 shows the five sheet designs (V1–V5). Each variant was run once, so a single run cannot separate a layout effect from run-to-run variation; for the central case we therefore repeated the judgement on the saved sheets (below).

Table 4. Luderick, IoU ≥ 0.3, same SAM tracks in all rows.

variant	audit evidence	recall	precision	exact MaxN	Σ|err|	queued
SAM alone	—	97.4%	78.7%	2/6	4	—
min length 4	no audit; tracks under 4 frames dropped (threshold chosen on these labels)	97.4%	79.2%	4/6	2	—
V1	one crop per track (its largest box); one crop per peak box	95.9%	79.3%	3/6	3	13/27
V2	three moments per track; peak box with the track before/after, judged frame in the middle	97.4%	79.2%	4/6	2	7/27
V3	V2 with 2-px borders and captions in the prompt	97.4%	79.2%	4/6	2	10/27
V4	V2, but when a track starts or ends at the judged frame, that frame moves to the strip's edge (no repeated crop)	96.3%	79.0%	**5/6**	**1**	9/27
V5	V2 layout restored; A2 and relabelling added (inactive on this task)	97.4%	79.2%	4/6	2	9/27
V6	V5, but A4 asks about the requested category ("fish"), not A1's species; A3 also checks pairs across words (none here)	97.4%	79.2%	4/6	2	9/27
final (frozen `cbaf59b`)	V6, with A4's example of another animal no longer naming the category, and the rules of §5 *Protocol* (replayed, $0)	97.4%	79.2%	4/6	2	9/27

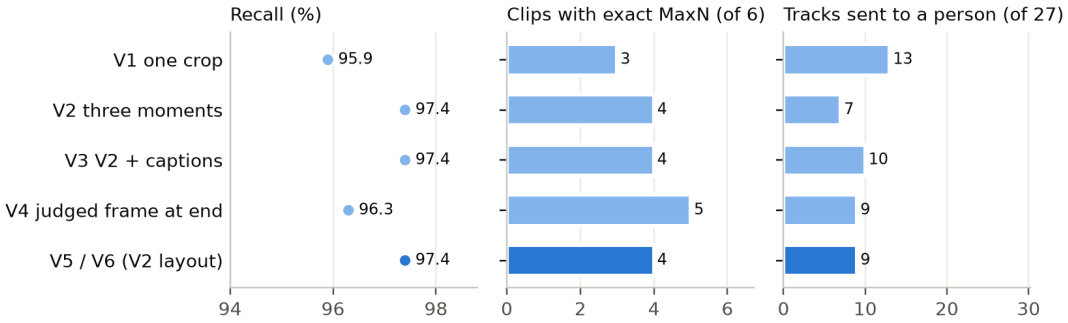


Figure 6. The same SAM tracks audited with five sheet designs (V1–V5; V6 uses the V5 sheets). Recall is drawn as dots because its axis does not start at zero. V4 has the best MaxN score and the worst recall of the three-moment designs; see text.

One crop is not enough. With a single crop per track (V1), A1 called a faint fish behind seagrass "only seagrass and water" and a 36-frame track left the count; A4 did the same to another faint fish in frame 35 of 20190618_9. Recall fell from 97.4% to 95.9%. With three moments (V2) the same fish are kept: the body that is a smudge in one crop is a moving fish across three.

Position within the strip matters, not only content. V4 changed one detail of A4: when a track starts in the judged frame, the strip showed that frame first and two later frames, instead of repeating it on the left. The faint fish in frame 35 of 20190618_9, whose track starts there, was again called "seagrass, no fish" and recall fell to 96.3% (Figure 7). Restoring the middle position (V5) restored 97.4%.

To check that this is not one unlucky call, we sent the saved A4 sheets for frame 35 byte for byte, 24 more times each without the cache: the V4 sheet and the V5 sheet, each with the prompt of the code version that produced it (both verified to hit the original cache entries), and the V5 sheet with V6's question (original call from run eval7), 72 calls in all. Together with the three original calls:

Table 4b. Spark's answer for the faint fish (tile D7) on the saved frame-35 sheet, 25 calls per layout. Data: `report/paper/data/repeat_peak_f35.json`; tally: `report/paper/scripts/repeat_peak_score.py`.

layout	called not a fish ("seagrass", "empty water")	called "other fish" (not the label)	kept
V4: judged frame at the strip's edge	11 of 25	0 of 25	14 of 25
V5: judged frame in the middle	0 of 25	4 of 25	21 of 25
V6: middle, question names the category ("fish")	0 of 25	4 of 25	21 of 25
final: V6 with the example "an eel or hagfish" instead of "an eel, hagfish or fish"	0 of 25	0 of 25	25 of 25

With the frame at the edge, Spark failed to see an animal in 11 of 25 calls; in the middle, never (Fisher exact test, two-sided $p = 0.0002$). These 25 + 25 include the two original calls that prompted the experiment; on the 24 + 24 repeats alone the split is 10 of 24 vs. 0 of 24 ($p = 0.0006$). On this sheet, then, the position of the frame changes whether the faint body is seen as a fish at all; one tile in one frame cannot say how general the effect is. Whether the track survives differs less (removed in 11 of 25 vs. 4 of 25, $p = 0.06$), because in the middle layout Spark sometimes recognised a fish but answered that it was "other fish" than the label being checked, which also removes it (§7.5). Asking about the requested category (V6) did not remove that answer (4 of 25); removing "fish" from the question's example of a different animal (final code) did (0 of 25; §7.1). The single runs of Table 4 thus each show one draw: V4's 96.3% is the outcome in 11 of 25 calls, and V5's and V6's 97.4% in 21 of 25 each. The direction agrees with reports that multi-image VLMs weigh inputs differently by position (Tian et al., 2025); their setting is separate images, ours is crops within one composite sheet, so we call the effect similar rather than the same.

A better MaxN for the wrong reason. V4 has the best MaxN score of all variants (exact on 5 of 6 clips). The improvement is on 20190618_9, where the human MaxN (9) is itself likely too low (the frame that sets the system's 10 holds two real fish without a box): by wrongly rejecting a real fish, V4 moves the peak count from 10 to 9 and "matches" the human number. A MaxN-only evaluation would have selected the variant that discards real animals. We therefore treat recall on human boxes as the constraint and MaxN agreement as the objective, and we checked every sheet or prompt change against the frame-35 case and against the s004 peak frame before accepting it, which is also why these clips are not a held-out test (§5).

The queue moves while the counts do not. V2, V3 and V5 count exactly the same boxes, yet send 7, 10 and 9 of the 27 tracks to a person. The differences come from confidences near the 0.7 threshold changing with cosmetic sheet changes (borders, captions) and between

runs. The queue size is therefore a noisy quantity at this sample size; we report the final run's 9 of 27 and the observed range of 7 to 10.

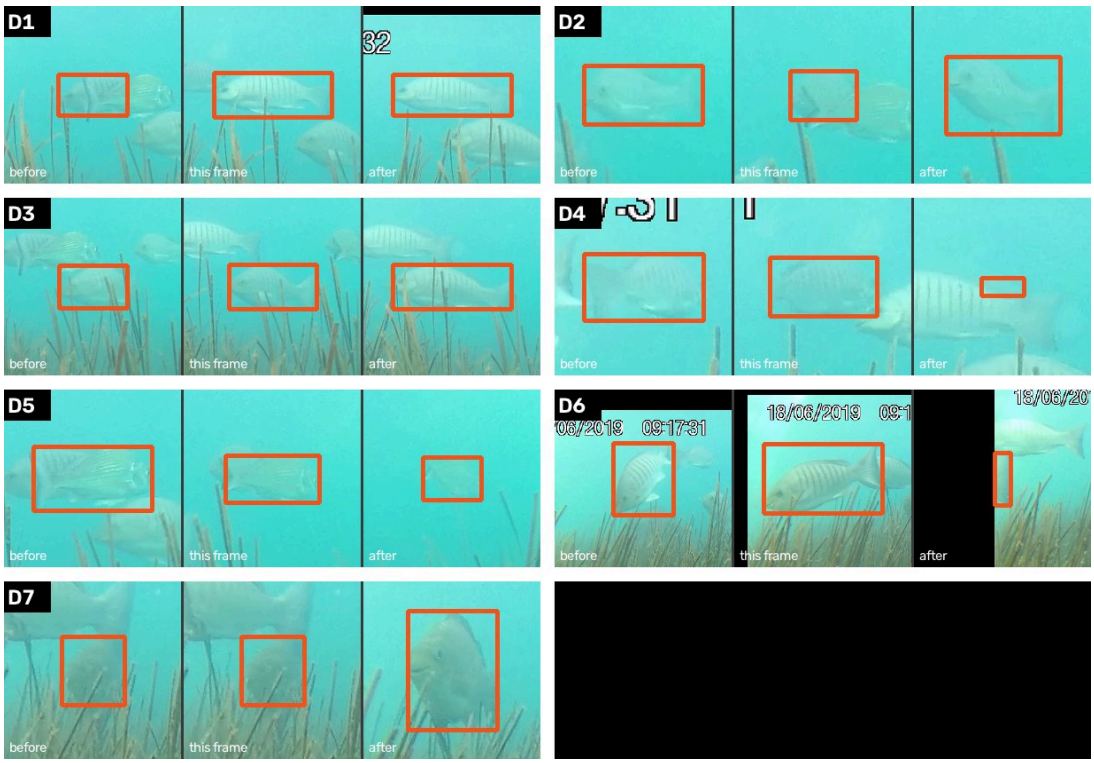


Figure 7. A4 sheet for the peak frame of 20190618_9 (V5). Each tile shows "this frame" in the middle with the same track before and after. The faint fish that V1 and V4 rejected is kept when its frame sits in the middle of the strip.

7.2 Sensitivity to the matching threshold

Table 5. Recall / precision of SAM alone and of the final pipeline at three IoU thresholds. MaxN columns do not depend on matching.

IoU	SAM alone	audited pipeline
0.3	97.4% / 78.7%	97.4% / 79.2%
0.5	96.7% / 78.1%	96.7% / 78.6%
0.7	85.5% / 69.0%	85.5% / 69.5%

The conclusions hold at IoU 0.5 (Table 5). At 0.7 both systems lose about 12 points of recall together, which reflects box geometry (SAM's mask-derived boxes vs. hand-drawn boxes on partly hidden fish) rather than detection; the audits neither help nor hurt localisation, as expected since they only keep or remove whole tracks.

7.3 Where the counts come from

Across the four discovery clips the pipeline produced 44 tracks after merging: 32 from discovered words and their zoom or reversed passes, 10 from motion box prompts and 2 from the requested word (s002's two "fish" tracks, both queued). Seven tracks were not counted (a confident A1 rejection, a low-confidence motion track, or an A2 "no"); A3 judged 9 nested pairs (8 in s004, 1 in s003), of which 6 were the same animal counted twice (all in s004,

where shrimp crowd; Figure 9) and 3 were kept apart; A4 checked 11 peak frames and rejected 3 boxes: a box on the trap's blue pipe (correct, checked by eye), an 11 × 15-pixel box in empty water (not checked), and the only box of the real nautilus in s002's frame 0, called "anemone tentacles" (wrong; the MaxN survived because a later frame became the peak, but the removal window erased the first arrival, reported at 1.2 s instead of 0 s). Of the 10 motion tracks, 4 were counted (s001 two nautiluses, s004 two shrimp). What discovery and the audits each contribute is separated in Table 3 and §6.3 (no taxon without discovery; names, the s003 false taxon and the s004 duplicates fixed by the audits). Two further components show only in this accounting and the development history:

- **Without the motion residual** the s001 nautiluses are never found on the original clip: every word, block and 3× cell returned 0 frames on this night clip (brightened, "shell" reaches one of them for 9 frames), so the taxon count would be 6 of 7, and s004 would lose two shrimp at the frame edge.
- **Without the second look (v13)**, s003 counted a "large striped shrimp" that does not exist (1 false taxon); with it (v14 onward), 0.

7.4 What the segmenter responds to

Table 6. Prompt probes on SAM 3.1 (frames tracked, or detections per frame, as stated). These are observations on our clips, not general claims about the model.

probe	result	consequence for the design
"animal" vs. "fish", all 39 clips	"animal": no box on 29 clips; on the 30 luderick clips at most 1 per frame where "fish" gave up to 10; it overlapped 3 of the 11 non-fish animals the pipeline counted (s002 nautilus and crab, DH2 crab). Data: `data/animal_probe/`	the VLM translates requests into basic-level nouns
"luderick" (the species' name), six development clips	0 detections on all six ("fish": a peak of 1 to 10 boxes per frame, by clip)	species come from the audit; SAM gets the plain noun
"hagfish" vs. "eel", s004	0 vs. 24 frames	word ladder; Spark's name is kept as the label
"shell" vs. "nautilus", s002–s003 (development spikes)	"shell" tracked the s002 nautilus (47 frames) but returned nothing on an s003 crop where "nautilus" tracked (development observation); in the final runs "nautilus" tracked 16–48 frames on three clips, and on s001 every word, "shell" included, returned 0	the VLM's own short name is tried first
forward vs. reversed, same word	nautilus 16 → 60 (s004), 33 → 60 (s003); shrimp 40 → 60 (s004)	every word that works is re-run on the reversed clip
full frame vs. 3× cell, s002 crustacean	"crab" 0 → 5, "shrimp" 0 → 12 frames	closer look in the cell the VLM names
box prompt with no word, s004 bottom-edge shrimp	57 of 60 frames in the first test (55 in the pipeline); no word, tile or reversed pass ever tracked it	motion residual → box prompt
box touching the frame edge	HTTP 400 ("lies outside the frame")	boxes kept two pixels inside

The pattern in Table 6 agrees with SAM 3's stated design, which constrains text to "simple noun phrases" for "atomic visual concepts" and notes weak zero-shot generalisation to fine-

grained concepts in niche domains (Carion et al., 2026), and with benchmark findings for open-vocabulary detectors (Yao et al., 2024; Nebbia and Kovashka, 2024; Parisot et al., 2023). We found no prior measurement of a hypernym returning nothing or of a rare-species name failing where a basic-level neighbour works for SAM-class models. SAM 3's ontology does include an "animal" category (Carion et al., 2026, App. D.2), so we state the result as counts on these 39 clips, not as a property of the model. The asymmetry between word and box prompts is consistent with Lian and Li (2024) for SAM 2 underwater.

7.5 Failure modes of the VLM auditor

Name anchoring. In s003, a motion box had drifted onto the legs of the trap's net. A1, whose context string listed the discovery step's (wrong) guess "large striped shrimp", passed the track as that animal with confidence 0.75. Shown the same track at other moments with no label and no names, Spark answered "looks like trap mesh, not animal" (Figure 8). The mechanism is the anchoring and misleading-cue effect reported for LLMs and MLLMs (Jones and Steinhardt, 2022; Dang et al., 2024; Lou and Sun, 2024); the consequence here is a false taxon in a count. Lou and Sun (2024) report that instructing a model to ignore an anchor does not remove the effect; we removed the anchor instead, which is the independence step of Chain-of-Verification (Dhuliawala et al., 2023).

Vocabulary priming. The motion prompt offered example nouns (shrimp, crab, fish, eel, worm, starfish, snail, octopus). In s001 Spark described a spot as a "striped nautilus shell" and named it "snail", a word from the list. We now let a name that the discovery step already gave win when Spark's own description uses it, and A1 relabels such tracks; the general lesson is that example lists in prompts act as answer options.

Layout and context sensitivity. Beyond the strip-position effect in §7.1, the same pair of nested masks (s004, frame 27) was judged "same animal" on one sheet and "different" on another, depending on which other pairs shared the sheet. Decisions that set a count are therefore made on small sheets with three-moment crops, and doubtful cases go to the queue rather than to a vote.

Nondeterminism and threshold noise. Identical counts with queues of 7, 10 and 9 tracks (§7.1) show confidence values moving across the 0.7 threshold between cosmetic variants and runs, and the identical frame-35 sheet received different verdicts across 25 calls (Table 4b), consistent with reports of nondeterministic outputs at fixed settings (Atil et al., 2024; He, 2025). Caching makes any given run exactly reproducible, but a fresh run can differ by a track, and by more when Spark's proposed words miss the one that works. In four empty-cache rehearsals through the web screen (request "fish", two takes per clip, $0.20–0.26 and 2–3 min each; `report/paper/data/rehearsal/`), DH11's amphipod was counted once (1) and queued once (0), and s003 counted 3 nautiluses in one take but 1 in the other: there Spark described "chambered nautilus cephalopod" and proposed only "cephalopod", "shell" and "snail", so "nautilus" was never tried and only a zoom tile's "cephalopod" track (40 frames) was counted. No take counted a non-animal.

Wrong rejections in the peak check. A4 can remove a real animal. In s002 it rejected the only nautilus box in frame 0 as "anemone tentacles"; the nautilus still counts because a later frame, where it is clearer, became the peak, but the ±1 s removal window moved its reported first arrival from 0 s to 1.2 s (frame 6 instead of 0), and a label with a single short appearance

would have been lost. And because the A4 question names the label, a real fish of another species can be rejected for not being that species: on the frame-35 sheet Spark answered “other fish” for a fish under the label “striped seabream” in 4 of 25 calls (Table 4b). Under the v16 rule such a track was removed from the peak frame and not moved to another label, so a per-species check silently lowered the total. V6 asks A4 about the requested category for word-prompted tracks and leaves the species to A1; every saved count is unchanged, but on the same sheet the fish was still called “other fish” in 4 of 25 calls (Table 4b). The question’s examples of a different animal read “an eel, hagfish or fish”, which names the category itself; with only that phrase changed to “an eel or hagfish”, the fish was kept in 25 of 25 calls on the same sheet (4 of 25 vs. 0 of 25, $p = 0.11$; `report/paper/scripts/repeat_peak_fixexample.py`, written before the calls). The final code uses that phrase, and an answer that names another animal of the same group (“other fish” under a fish label) no longer removes the box: it stays in the count and the track goes to the queue for its species. An answer naming a part, a duplicate or no animal (“tail of another fish”, “sediment between crab legs”) still removes it.

Malformed output. During development Spark twice returned replies that were not valid JSON (for example `{ polo":1 }:`); because one was cached, every replay failed. A reply that fails its schema check is now never cached and is asked once more. Since bad replies are no longer stored, the frequency cannot be recomputed from the cache; we report this as an engineering observation.

Species names are proposals. On the luderick clips, A1 labelled 4 tracks “luderick”, 8 “striped seabream” (a different species), 13 remained “fish (unidentified)” and 2 were rejected. Evaluation against boxes ignores labels, but a per-species MaxN reported to a user would split one species into two names (in 20190618_9: “striped seabream” 7 and “fish (unidentified)” 3). Species-level labels from a general VLM need the human review the queue provides, and a taxonomic list for the survey region would be a natural constraint to add.

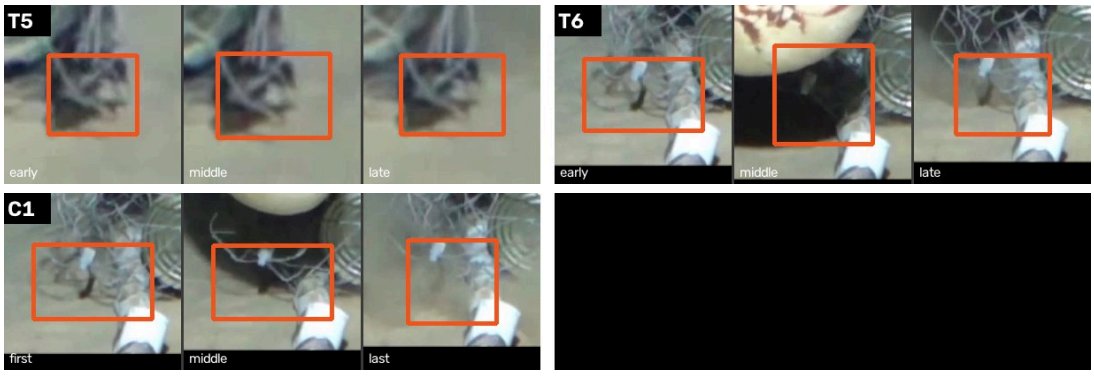


Figure 8. Name anchoring, s003. Top: A1 sheet; the right group (T6) is a motion box on the trap’s net legs that A1 passed as “large striped shrimp” (0.75), a name listed in its context. Bottom: the second look at other moments, no label, no names: “looks like trap mesh, not animal”. The track is not counted.

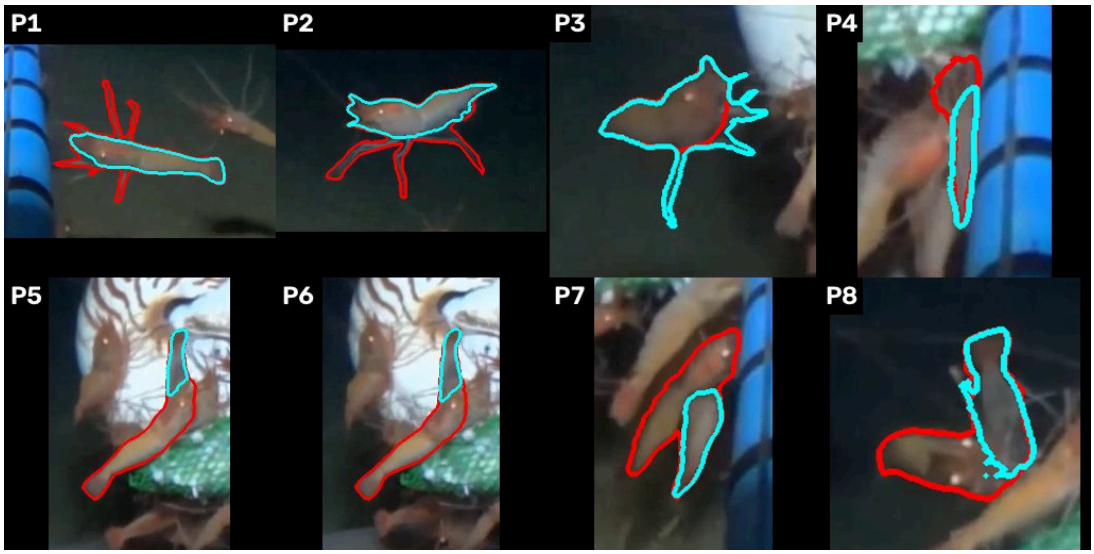


Figure 9. A3 sheet, s004: nested masks drawn red and cyan; Spark judged six of eight pairs to be one shrimp counted twice (for example body only vs. body with legs).

7.6 Swapping the VLM: Muse Wonder on the same questions

Meta's API offers no Llama vision model (`GET /v1/models`, 3 Nov 2026: Muse Wonder, SAM 3.1, Muse Spark 1.1–1.3, Voice Transcribe, Image), so the only other model on it that reads images is Muse Wonder, Meta's largest, priced at $4.00 / $20.00 per million input / output tokens against Spark's $1.25 / $4.25. We asked whether the larger model would change what the pipeline concludes. The protocol was committed before any Wonder reply was seen (`report/paper/vlm_swap_protocol.md`). Each of the 15 discovery clips' final runs was replayed offline with Spark's cached answers driving every step and no SAM call; each of the 101 questions Spark answered (audit 30, peak check 29, discovery 15, motion look 14, motion confirm 9, pair 4) was sent unchanged, with the same images, settings and reply check, to `muse-wonder-1.0`. This is a same-question comparison, not Wonder's own end-to-end run: SAM never tracked Wonder's names, and the later questions are the ones Spark's answers led to.

Table 7. Muse Spark 1.3 and Muse Wonder 1.0 on the same 101 questions. Discovery names were graded blind by a separate grader (a Claude Sonnet subagent, read-only) that saw both lists per clip in random order without the model names. Data and scorer: `report/paper/data/vlm_swap/`, `scripts/vlm_swap_score.py`.

measure	Spark 1.3	Wonder 1.0
Held-out DH1–DH11: marked non-fish animals named, strict (crab and shrimp/lobster separate)	5 of 5	4 of 5 (DH3 crab named only "squat lobster")
same, lenient (one crustacean group)	5 of 5	5 of 5
Development s001–s004: taxa named (of the seven in Table 3)	7 of 7	7 of 7
Names supported by no reference list, 15 clips	5	12, of them 4 not animals (bait carcass, bacterial mat ×2, seaweed)
Rows a person judged "animal" (metric 3, 7 rows): rows with tiles the model rejects	— (all 7 counted from Spark's answers)	3 (DH3 spider crab on 4 audit and 2 peak-check tiles; DH2 crab and sea anemone, 3 shared audit tiles)
Cost per call, the 87 questions whose Spark reply stored its usage (the 14 motion looks did not)	$0.0047	$0.020, about 4.3× ($0.023 over all 101)
Seconds per call	not logged	4.7 mean, 61.6 max

Tile by tile, Wonder agreed with Spark on 106 of 121 audit tiles, 71 of 77 peak-check tiles, 12 of 15 motion confirmations, 88 of 111 motion-look tiles and 11 of 12 same-animal pair verdicts; where they differed, Wonder said "no" to a tile Spark kept in 38 cases and "yes" to one Spark dropped in 10. This compares the per-tile yes/no field only; the protocol said the pipeline's own parsing would be used, which also applies confidence thresholds, and we did not apply them. The first attempt lost 26 Wonder calls to HTTP 503 "service overloaded" and 16 more to our own error: the spending cap is on the project's whole ledger, and an hour-long run (§6.4) was spending at the same time. The second attempt retried 503s, which are transport errors rather than replies, and finished all 101 with no unreadable reply; the first attempt's calls are kept. Wonder cost $2.33 (Spark's cost on the same questions: `scripts/vlm_swap_spark_cost.py`, $0 to rerun).

On these questions the larger model is more conservative where it matters for counting (it rejects tiles of an animal a person was certain of) and less careful where it matters for discovery (more names with nothing behind them, four of them not animals), at about 4.3 times the cost per call. Nothing here argues for replacing Spark, and because Spark's answers chose the questions, it does not show what a pipeline built around Wonder would count.

7.7 The plain SAM 3 Agent pattern as a discovery baseline

SAM 3's "SAM 3 Agent" lets an MLLM propose noun phrases, look at the returned masks and revise (Carion et al., 2026, §6). To test whether that pattern alone does what our discovery steps do, we ran it with our models on the 15 discovery clips (protocol committed before any call: `report/paper/sam_agent_protocol.md`): the final run's own discovery reply proposes up to four things with their first words; SAM 3.1 tracks each phrase on the full frame, forward only; one Spark call per round sees the contact sheet and the three frames with most boxes for each phrase, and answers keep, revise (one new plain noun) or drop; at most three SAM rounds per thing; no zoom, reversed pass, motion residual or audit. A separate grader scored the kept things and the pipeline's counted labels blind against the reference lists.

Table 8. Pipeline vs. a propose–segment–look–revise loop on the same 15 clips. Data: `report/paper/data/sam_agent/`.

	pipeline (counted)	**loop (kept)**
Held-out DH1–DH11: marked animals reached, strict / lenient	4 of 5 / 4 of 4	3 of 5 / 3 of 4
Development s001–s004: taxa reached (Table 3)	7 of 7	5 of 7
Entries supported by no reference list	1 (DH7 crab)	1 (DH7 crab)
Things revised / kept after a revision	—	16 of 25 / 2
Cost	Table 3b, §6.4	$0.41 (41 of 56 SAM calls already cached)

The loop missed the DH3 spider crab (“crab”, “spider”, “crayfish”: 0 frames on the full frame), the s002 red crab (“crab”, “lobster”, “bug”: 0) and the s003 nautilus (“shell”, “snail”, “octopus”: 0), and found one of the four s001 nautiluses (“shell”, 9 frames). Each miss is an animal the pipeline reaches by another route: “crab” or “shrimp” in a 3× cell for the two crabs, the listed word “nautilus” (33 frames) for s003 (in one of two empty-cache rehearsals the pipeline’s own Spark did not list it either and counted 1 of 3, §7.5), motion box prompts for s001. Revisions rarely helped (2 of 16 revised things were kept, as “coral” for a sea anemone and “bug” for an amphipod) and in s003 moved away from the word that works. Like the pipeline, the loop kept nothing that is not an animal. The blind sheet’s pipeline list for s004 lacked “hagfish” because a shared helper drops labels of the fish group; the pipeline counts it (Table 2). This supports positioning the pipeline as an application of the known pattern: the pattern supplies the naming, and the counts depend on the extra routes to the segmenter.

7.8 An open-vocabulary detector as the discovery step

An open-vocabulary detector is the obvious way to look for animals without a VLM. We fixed a protocol before running one (`report/paper/open_detector_protocol.md`): OWLv2 (Minderer et al., 2023; `google/owlv2-base-patch16-ensemble`, CPU, no fine-tuning) on every frame of the 15 discovery clips that the final runs tracked, with 17 queries (“animal” plus the 16 fixed words of §6.3), score threshold 0.2 (0.1 and 0.3 as sensitivity). The reference is the 45 non-fish tracks the pipeline counted; a track is found when some detector box overlaps it (IoU ≥ 0.3) in at least 3 frames.

Table 9. OWLv2 on the 15 discovery clips (900 frames), against the 45 counted non-fish tracks.

threshold	**tracks overlapped**	**only through “fish”**	**through a non-fish word**	**right-group word**	**only through “animal”**	**clips / frames with an “animal” box**	**non-fish boxes off every pipeline track**
0.1	39	14	24	12	0	5 / 40	6,107 of 8,123
0.2	**39**	**17**	**21**	**11**	**0**	**3 / 30**	**1,716 of 2,903**
0.3	37	19	18	10	0	2 / 22	685 of 1,530

The detector puts a box on most of what the pipeline counted, but it does not name it: "animal" never accounts for an overlap and fires in 30 of 900 frames, 17 tracks are boxed only as "fish" (18 of the 19 s004 shrimp tracks are overlapped, none by a crustacean word), and only 11 carry a word of the animal's group. The misses are five DH7 crab tracks (the cold-seep time-lapse) and one s004 shrimp track. As a discovery step it would leave a person 1,716 unnamed non-fish boxes (about two per frame) that match no pipeline track; whether any of them is an animal the pipeline missed is not judged here. It ran at 14.9 s per frame on a shared 6-core CPU. Added after the protocol: s004's counted "hagfish", which falls outside the reference because its name maps to the fish group, is overlapped in 29 of its frames. Outputs and per-clip rows: `report/paper/data/open_detector/`. One detector at base size is a limit: a larger detector or Grounding DINO might name more, and measure 1 measures agreement with the pipeline, not recall.

8 Scope

Counting was measured on 30 luderick clips from one estuary, discovery on 15 12-s clips from nine studies (4 nautilus development clips, 5 and then 6 held-out clips); every number above is for these clips. Counts are what a reviewer confirms from the queue and the evidence frames, and species names are the VLM's proposals (§7.5). Videos over 10 minutes are split into segments and merged (§6.4); one 53-min deployment has been run end to end ($15.08 and 3.5 h in its completed attempt), without ground truth, so counts on long videos are measured for cost and robustness only.

9 Ethics and broader impact

The intended use is to reduce the hours ecologists spend watching survey footage, while keeping every number traceable to a frame and every doubtful track in front of a person. The main risk is misplaced trust: an automated MaxN can be wrong for reasons that are invisible in a table (a wrongly rejected faint animal, a split species name), and survey counts feed management decisions. We mitigate this by reporting the queue as part of the output and by publishing failure cases alongside successes. The footage contains no people except the first two minutes of the NOAA recording, where crew on the boat deck are visible (mostly torsos and legs); the pipeline counts them as "human"; we do not show those frames in the paper, figures or demo, though the reply cache in the repository keeps the crops Spark was shown. We used only openly published research footage, under CC BY licences or, for the NOAA recording, as US federal government work (Data statement). The discovery reference lists were made by people who are not marine biologists, as stated where they are used.

10 Reproducibility

The repository (github.com/kitavkdl/reefcensus, MIT licence; the submitted version is tagged) contains the code, the prompts, the four final discovery runs with every sheet Spark judged (`demo/runs`), and a scoring bundle in `report/paper/data/`: the human boxes of the six luderick clips (from Ditria et al., 2021, CC BY 4.0), SAM 3.1's reply for "fish" on each clip, the `result.json` of every scored luderick run (V1–V6 and the merge-only baseline), the held-out runs (§6.2), the no-audit discovery runs, the held-out discovery runs with both reference lists,

the circled sheets and the label-free judgement (`data/discovery_heldout`), the two frame-35 sheets, the replies of the repeated calls, the per-frame replies of the Spark-alone baseline (`data/spark_count`), the fixed-vocabulary tracks (`data/fixed_vocab`) and a copy of the cost ledger. From a fresh clone, with no data download and no API key (Python 3 with NumPy, OpenCV and, for the figure, matplotlib),

```
python3 report/paper/scripts/rescore.py              # Tables 1, 1b, 4, 5: baselines, queue
bounds, intervals
python3 report/paper/scripts/rescore_heldout.py      # Table 1c: held-out clips, MaxN range
python3 report/paper/scripts/spark_count_rescore.py  # Table 1c Spark rows, per-frame agreement
and MAE
python3 report/paper/scripts/score_discovery_heldout.py  # Table 3b: held-out discovery rows,
queue, cost
python3 report/paper/scripts/ledger_summary.py       # cost paragraph (§6.4)
python3 report/paper/scripts/fig_ablation.py         # Figure 6
```

reproduces the counting and cost numbers of §6.1, §6.2, §6.4 and §7.1–7.2 and Table 3b; we checked this in a clone with the live run files and video paths absent. Re-running the pipeline itself needs the source videos (luderick from PANGAEA, nautilus from the PLOS ONE supplement) and either the model-reply cache of our runs (`cache/`, 12 MB, in the repository) for a free, exact replay, or an API key and a spend cap for a live run, which can differ from ours (§7.5):

```
python3 fishcount/evaluate.py                                    # SAM "fish" alone vs. luderick
boxes
python3 fishcount/evaluate_pipeline.py --tag eval6               # final pipeline vs. the same
boxes (add --run --no-clean --no-refill to re-make the runs)
python3 fishcount/pipeline.py <clip> --no-scout --prompts fish --discover --zoom --no-clean --
no-refill   # discovery run (Table 2)
python3 report/paper/scripts/discovery_noaudit.py                # Table 3: no-audit discovery re-
plays (needs cache)
python3 report/paper/scripts/repeat_peak.py --check              # Table 4b: saved sheets hit the
original cache
```

Defaults added after the freeze. Every number above is from the code frozen at `cbaf59b`. Two track-hygiene steps were turned on by default afterwards (11/3), for crowded shallow-water video: *clean* (SAM ids reused for another object are split off, ids SAM shuffled within a frame are given back, and a dark counted track with no highlight gets a wide-view “animal or shadow” look) and *refill* (a SAM pass that used all 16 of its object ids is re-run from the frame where it ran out). `--no-clean --no-refill` restores the frozen behaviour (the discovery command above carries it). We re-ran every frozen evaluation once with the new defaults (tag `cr3`, log `report/paper/data/run_cr3.log`). Luderick is unchanged: development 97.4% recall, 79.2% precision, Σ|err| 2; held-out 98.9%, 77.2%, Σ|err| 12. Of the 15 discovery runs, 10 count exactly the same rows, the totals per animal group move only on DH8, and the discovery scores do not change (development 7/7, DH1–DH5 3 of 4 counted, DH6–DH11 1/1). The five that move: s004 caridean shrimp 12 + 3 unidentified → 14 + 2 (one fewer duplicate merge puts a 44-frame shrimp track under the species name; the shrimp group’s MaxN is 15 in both); DH5 fish 8 + pink cusk-eel 1 → fish (unidentified) 9 (same total; the split-off one-frame tracks enter the audit sheet and Spark names no species); DH6 leopard shark → tiger shark, which is what the video shows, plus one fish (unidentified) from a one-frame track Spark judged at 0.35;

DH7 crab 3 plus a separate “crab (unidentified)” 1 (two short tracks Spark called crab at 0.6; the crab group’s MaxN stays 3); DH8 fish 15 → 17 (fish group 16 → 18). Between the first and the last re-run with the new defaults, the id repair alone moved DH8 from 18 to 17: the one-frame pieces it splits off are rejected by the audit, which takes one box out of the peak frame. One scoring change was made after a re-run: in the first one (`DH5_cr`) the audit named DH5’s cusk-eel “new zealand ling”, the species the frozen run had called pink cusk-eel, and since no word group knew “ling” the scorer counted a fish as a non-fish discovery; `score_discovery_heldout.py` now treats “ling” as a fish name. No frozen run and no `cr3` run contains the word, so no reported score depends on it.

The repeated calls of Table 4b are live calls; their replies are stored in `report/paper/data/repeat_peak_f35.json`. All calls used the model IDs fixed in `fishcount/api.py` (`sam-3.1`, `muse-spark-1.3`), and the ledger records the time of every paid call (October–November 2026). Hyperparameters are listed in Appendix B.

11 Data statement

- **Luderick:** Ditria, E.M., Connolly, R.M., Jinks, E.L., Lopez-Marcano, S. (2021). Annotated video footage for automated identification and counting of fish in unconstrained marine environments. PANGAEA, doi:10.1594/PANGAEA.926930. CC BY 4.0. Data report: *Frontiers in Marine Science* 8:629485.
- **Nautilus:** supplementary videos S1–S4 of Barord et al. (2014), *PLoS ONE* 9(6):e100799, CC BY.
- **Held-out discovery:** Wikimedia Commons copies of PLOS ONE supplementary videos: S4, S1 and S3 of Anderson & Bell
2014. (DH1–DH3) and S2 of Anderson & Bell (2016) (DH4), CC BY 4.0; Video S1 of Zintzen et al. (2012) (DH5), CC BY 2.5. URLs and SHA-256 hashes are in `report/paper/data/discovery_heldout/clips.json`.
- **Second held-out discovery set:** Wikimedia Commons copies of PLOS ONE supplementary videos: Video S2 of Werry et al. (2014) (DH6), CC BY 4.0; Video S1 of Niemann et al. (2013) (DH7), Video S4 of Oliver et al. (2011) (DH8) and Videos S4 and S1 of Sturdivant, Diaz & Cutter (2012) (DH9, DH10), CC BY 2.5; Video S4 of Cuvelier et al. (2014) (DH11), CC BY 4.0. URLs are in `report/paper/data/discovery_heldout2/clips.json`.
- **Long video (§6.4):** NOAA Fisheries, Southeast Fisheries Science Center, reef fish baited video, station `T60233001`, camera A, from NOAA’s Open Data Dissemination bucket `nmfs_odp_sefsc` (Gulf of Mexico reef fish videos). US federal government work (public domain, 17 U.S.C. § 105); the files carry no per-file licence text, and NOAA asks for attribution to the Southeast Fisheries Science Center. Position, depth and date are not stated in the folder. Download script and SHA-256 hashes: `report/paper/data/long_video/`.

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Appendix A: prompts

The prompts are reproduced from `fishcount/pipeline.py` at the final version; `{…}` marks values filled in at run time. Comments in the source record the case that motivated each sentence.

Discovery. > You are helping a marine ecologist review underwater survey video. They asked only to count: {targets}. The image > is a contact sheet of {n} frames from one clip (frame index in each top-left label). Look for every OTHER kind of > living thing that the request did not ask about, including small, partly hidden or unusual ones. A segmentation > model will search the video for each thing, but it only understands plain everyday English nouns (like “shell”, > “shrimp”, “crab”, “jellyfish”, “worm”, “starfish”, “eel”); it returns nothing for species names, colours, or vague > words like “animal” or “creature”. Also report indirect signs that an animal may be present but hard to see: sand > or sediment puffs, a moving shadow, bubbles, bait being disturbed, a burrow opening. Return ONLY JSON: {schema with > `what`, `why_notable`, three `search_words`, `frames`, `approx_max_count`, `cells`; and `signs`}. At most 4 others > (most notable first) and 3 signs.

A1 Track audit. > You are checking detections from an automatic tracker in underwater survey video. Each tile is one tracked object, > labelled with its track id (e.g. T3), shown at three moments of the clip (early, middle, late; same zoom, the > orange box marks the object); the detector was asked to find “{label}”. Context from an earlier look at the clip: > {context}. For every tile return ONLY a JSON list: [{“track”, “is_target”, “species”, “confidence”, “reason”}]. > is_target = the tile really shows a whole or partial {target} (not seagrass, rock, bubbles, debris, a shadow, or a > duplicate fragment you cannot tell apart). Use all three crops: an animal keeps its body shape while it moves; weed > sways in place and sediment changes shape. A hazy or faint {target} seen through water or seagrass is still a > {target}. Be honest about uncertainty: blurry tiles should get low confidence.

A2 Second look. > Each group (C1, C2, …) follows one tracked object in underwater survey video at three moments (captions under the > crops), same zoom; the orange rectangle marks the object. The tracker was started from a moving spot, so it may have > locked onto something that is not an animal. For every group return ONLY a JSON list: [{“tile”, “is_animal”, > “name”, “confidence”, “reason”}]. is_animal = the rectangle holds the same living animal in the crops (a whole one or > most of one; small, pale or blurry is fine). Answer false when it holds a part of a trap, net, mesh, rope, cans, > bait, rock, seagrass, a shadow, sediment or empty water, or only a tentacle, fin or tail of a bigger animal whose > body is elsewhere. Things fixed to the trap or seabed keep the same place against them in every crop; an animal > changes place or pose. Be honest: if you cannot see an animal, answer false.

A3 Same animal. > Each tile is a crop from one frame of underwater survey video with two outlines drawn on it: a RED outline and a > CYAN outline, made by an automatic tracker. […] same_animal = true when the cyan outline marks the same single > animal as the red one (or a part of it, for example its body without the legs or its head). false when the cyan > outline marks a different individual, for example a second animal next to, behind or in front of the first, even if > the red outline also spills over it.

A4 Peak-frame check. > Each tile is one object that an automatic tracker counted as "{name}" in a single frame of underwater survey video; > this frame sets the maximum count, so every tile matters. […] Each tile shows that frame in the middle ("this > frame", orange box) and the same tracked object about half a second before and after (left, right) at the same > zoom. […] is_target = the middle crop mainly shows a whole or partial {name}. Answer false when the box is on a > different animal (for example part of an eel, hagfish or fish), a net, rope, pipe, rock, sediment or empty water. > Use the side crops: an animal keeps its body shape while it moves; weed sways in place. If it is too blurry to > tell, answer true and say "unclear".

Motion spots. > Each group (M1, M2, …) shows one spot of underwater survey video three times, left to right: early, middle and > late in the clip, same zoom, following the spot. The orange rectangle marks where something moved and no detector > has boxed anything yet. […] Compare the three crops: an animal keeps about the same shape and size (it may turn or > move); a sand or sediment cloud spreads, thins or changes shape from crop to crop, and a shadow changes with the > light. Black areas are outside the picture: an animal cut off by the edge of the picture still counts. […]

Appendix B: hyperparameters

component	parameter	value
ingest	frame rate; max width; segment length (longer videos split, final)	5 fps; 1920 px; ≤ 600 s
low light (final)	dark if Lab L median and mean below; then	15 and 35 (set from the development clips, post hoc); gamma 0.5 + CLAHE clip 2.0
clean (default after the freeze; `--no-clean`)	ID swap undone when an ID jumps by more than / the handed-over box sits on the other ID's predicted spot within; a track is cut when gone more than and back away from its last box; the audit's largest-box moment skips boxes touching the frame edge	2.0 box sides / 0.6 side; 10 frames
shadow look (with clean)	counted track darker than the ring around it and without a highlight: box median L − ring median, box 95th-percentile L − ring median; rigid companion above: offset spread, mean distance, height above, size ratio	< −10, ≤ 10; 0.12, ≤ 2.0, ≥ 0.2 sizes, 0.6–1.6
refill (default after the freeze; `--no-refill`)	a SAM pass that used all 16 IDs is sent again from that frame: window, overlap, calls per window of frames; inside the per-video cap	100 frames, 2 frames, 4
zoom	block (2/3 of frame) upscale; cell (1/3) upscale	1.5×; 3×
discovery	items per clip; words per item; zoom tiles	≤3; ≤3; ≤3
motion	analysis scale; threshold; temporal check; blob area; min path length; spots shown; boxes prompted; Spark confidence	0.5; median + 6·MAD (≥ 8, CIELAB); frame t±2 differ by > 0.8 × threshold; 12–2,500 px (analysis scale); 6 frames; ≤8; ≤5; ≥0.6
motion	counted only if	A1 ≥ 0.7 accept and A2 yes (else queued, not counted)
motion	box growth; grid; edge margin	+40% (≥ 48 px); 8 px; 2 px
merge	overlap to drop (same label); alias if duplicate on	IoU ≥ 0.5 for uncut zoom detections, else max(IoU, containment − 0.2) ≥ 0.5; ≥ 50% of frames
merge	motion track absorbed into a word track	IoU ≥ 0.3 in ≥ 3 frames
stitch	max gap; min IoU	10 frames; 0.2
A1	tracks per sheet; crops; label / reject confidence	12; largest-box frame ± ⌊length/3⌋; ≥ 0.7
Spark	reasoning effort; max tokens	low; 6000
A2	crops along path; animal confidence	1/8, 1/2, 7/8; ≥ 0.5
A3	nesting (mask containment), same word; across two words (final); pairs per sheet	≥ 0.8; ≥ 0.6; 16
A4	side gap; removal window; rounds; tiles per sheet	3 frames; ±5 frames (±1 s at 5 fps); 3; 12
evaluation	IoU match; MaxN frames	0.3 (also 0.5, 0.7); annotated frames only